

CHAPTER 1

1.1 Introduction

In this highly developing society time and man power are critical constraints for completion of task in large scales. The automation is playing important role to save human efforts in most of the regular and frequently carried works. One of the major and most commonly performed works is picking and placing of jobs from source to destination.

The pick and place robot is a microcontroller based mechatronic system that detects the object, picks that object from source location and places at desired location.

The main exploration in this project is about robotics project including a description of the recently programmed behaviour. The reports begin with an overview of the robot integrated system. It then continues with a description of the aspects of the robot; including robotic arm mechanisms their motor and servo actuators.

Robots are indispensable in many manufacturing industries. The reason is that the price per hour to control an automaton may be a fraction of the price of the human labour required to perform a similar operation. More than this, once programmed, robots repeatedly perform functions with a high accuracy that surpasses that of the foremost practiced human operator.

Human operators are, however, far more versatile. Humans can switch job tasks easily. Robots are unit engineered and programmed to be job specific. Today's most advanced industrial robots can before long become dinosaurs. Robots are unit within the infancy stage of their evolution. As robots evolve, they're going to become a lot of versatile, emulating the human capability and skill to modify job tasks simply.

This complete system allows the robot wander about its environment and to interact with certain objects that it encounters. The sensors provide the robot with its only information about the environment that it inhabits. These sensors include IR sensors on the robot platform which collect data while the programmed behaviour translates the

information into commands to its actuators. The robot's behaviour include line following, pick the object up, and deliver them to special areas in its environment.

1.1.1 Background

Pick & Place robots are used in a wide variety of material transfer applications. Basically, the machine takes a product from one spot in the manufacturing process and places it into another location. A good example is a robot picking items of a conveyor belt and placing them into packaging boxes.

The typical pick and place application requires high amounts of repetitive motion. Robots can eliminate human operation of hazardous tasks such as chemical spraying or heavy lifting. Pick and place robots have high return on investment when consistent shaped parts or containers are handled. Unlike human operators, robots also have the ability to work for an extended time.

The projected methodology permits a non-skilful person to mechanically link with a variability of engineering robotic apparatus. This project have been achieved which are developing the hardware and software for wireless mobile robotic arm, implementing the pick and place system operation and also testing the automaton that meets the standards of purpose project. From the analysis that has been made, it is clearly shows that its movement is precise, accurate, and easy to control and user friendly. The mobile robot has been developed successfully as the movement of the automaton as well as mobile and arm automaton will be controlled wirelessly. This robot is expected to overcome the problem such as placing or picking object that away from the user, pick and place hazardous object in the fastest and easiest way.



Figure 1.1 Typical Pick & Place Robot

1.1.2 Project Aim & Objectives

The aim of this project is to design an autonomous robot with complete system allows the robot wander about its environment and to interact with certain objects that it encounters. In order to achieve the aim of this project, several objectives are needed to be completed

- To study the concept on how pick and place robot function and operate.
- To design the hardware for pick and place robot.
- To make the system run continuously with less rest
- To make an interface between the programming of robot arm and the sensors.
- To make the robot's gripper has the function of pick and place.
- To make the robot capable to stop at the desire location accurately.

1.1.3 Scope

To implement an autonomous mobile robot which drive a vehicle without any human interface. It build by micro-controller (Basic Stamp), servo motors and infrared sensor and all the program is up loaded into the controller to makes its own decision and determine the position.

In order to design successful pick and places robot, scopes are required to assist and guide the development of the project. The scope should be identified and planned to achieve the objective of the project successfully on the time. The scopes for this project are:

- To design a program that controls the robot movement
- To design mechanical structure for the robot
- To fabricate Circuit Board for the controller

1.1.4 Benefit of Project

- It can show about the application among the PLC.
- Prevent back pain/back injuries due to picking up heavy loads by the operator.
- Increase productivity and efficiency.
- Eliminate the usage of human power to pick up things.

1.1.5 Importance of Work

In this scenario, the industries having a problem by risking human life in some hazardous duty service. Robots can work in environments so hazardous that an unprotected human would quickly die. Take for example bomb disposal. Typically these robots resemble small armoured tanks and are guided remotely by using cameras attached to the front of the robot. Robots can grab a suspected bomb and place it in an explosion proof safe box for detonation and/or disposal.

1.1.6 Problem Statement

This picks and place robot being designed to ease the sorting process of heavy materials. Usually the transfer process of the heavy materials is being carried out using man power and if the transfer process is repeated for a period of time, it can cause injuries to the operator. By using this particular robot, the operator will no longer have to bent and lift up heavy loads thus preventing injuries and increasing the efficiency of the work. Operator will make mistakes whether small or big everyone in a while .In the industrial world, the industry cannot afford to take kind of mistakes. Every mistake is costly whether in time, money, and material.

1.1.7 Report Outline

This project report has 8 chapters altogether. Chapter 1 gives some introduction and the objectives of this project. The literature review of this project is reported in chapter 2. This chapter reviews the related work that has been done by other people. Chapter 3 gives the overview behaviour of the robot. The main parts behind the working of the robot are explained in Chapter 4. It is the main part of this report. It has three main sections which are:

- Microcontrollers
- Motors
- Other equipment in the circuit

Chapter 5 is the Circuit Design of the project. Last but not least, chapter 8 gives the conclusion and future scope of this report.

1.1.8 Literature Review

The Pick and Place Mobile Robot has been attempted before in many laboratories and other places to upgrade the autonomous robot and the gripping system i.e. robot into a faster movement and more up to date controller. This chapter reviews existing techniques and ways of intelligent autonomous pick and place robot.

1.1.9 An Autonomous Self-Replicating Robotic System

The planet corporation in 1959 introduced a pick and place robot. In 1961, the rest industrial robot was commercialized by Animation Inc. Microprocessor technology was brought by INTEL in 1961. The real robot development process continued between 1968 and 1982 when various models of robots were developed by leading robot scientists in different universities, national laboratories and different industrial houses in the USA, Japan, France, UK, and other European countries.

Andrew B. Cushing, JackritSuthakom and Gregory S.Chirrikjian [1] from Department of Mechanical Engineering, Johns Hopkins University Baltimore, Maryland, implement the "An Autonomous Self-Replicating Robotic System". This robot consists with four subsystems: controller, left tread, right tread, and gripper/sensor subsystems. All subsystems are connected to others using magnets and shape constraints. The two light sensors in its navigation system to detect objects and also to track lines.

In this design and descriptions, LEGO RCX is used for controller the subsystem which t inside a chassis. The chassis's sides are used to connect to the left and right treads. The gripper/sensor subsystem is comprised of 9v LEGO DC motor, set of rack and

pinion gears used to drive the left/right of the gripper, a set of electrical connections and two light sensors (one is pointed downward, other the other is pointed forward).

In control and programming shows the grasping process consists of an aligning push towards the subsystem, and closing the gripper to grasp the subsystem. On the other hand, the assembly process consists of the opening the gripper to release the subsystem, and an aligning push forward to snap the subsystem to the controller. The replication process takes two minutes and fifteen seconds per cycle. Although each subsystem is required to place in its starting location, errors in initial position and orientation are not very critical. There were slight errors during the grasping process in a few experiments caused by improper placement of the subsystems.

1.1.10 Distributed Sensing and Control Framework for Mobile Robot

Another example, Chris Nowak, Rajaey Kased, Chin Pei Tang from Department of Mechanical and Aerospace Engineering, MAE405/405: Mechatronics, 2002 implement a project on "Distributed Sensing and Control Framework for Mobile Robot". The main task of this project is to transport a package from a pickup station to a drop-o station. The package transportation is done by wheeled mobile robot (WMR) with a gripper system in front of it, which works as a lifting device. The package availability on pickup station is done by a pushbutton on the platform of the station. Also, there is another pushbutton set up at the drop-o station to check whether the package has successfully sent.

To achieve this control scheme, they having a mobile unit and static unit, wireless communication is more adequate. Infrared and radio frequency communications are chosen as theirs communications scheme. These communication schemes are sending and receiving data with serial communication. WMR setup by servomotors: movement of the MMR, IR ranging system: distance detector, Fire-Stick II (IR transmitter): transmit data to base station, RF receiver: receive data from base station, Gripper system: the lifting device, Counterweight: to balance the moment of the WMR.

The basic directional motion required for this project is forward, backward, left and right. The whole motion is pre-programmed in the Basic Stamp on the WMR. The motion is done in two modes: calibration (open loop) and measurement (closed loop).The mode at each path is selected by trial-and-error.

Since calibration and trial-and-error might not work well all the time, they tried to use this ranging sensor to improve the WMR motion control.

1.1.11 The Development and Energetics of Slug Bot, a Robot Predator.

Ian Kelly, Chris Melhuish and Owen Holland, from Faculty of Engineering, University of the West of England, Bristol, BS16lQY, UK, implement "The Development and Energetics of Slug Bot, a Robot Predator". This robot capable of autonomous action on agricultural land and the robot will sustain its self by hunting and catching slugs.

In the design, the sensor used for detecting slugs, and the gripper used for catching them. This both parts are located at the end of a long articulated arm. During scanning the sensor and gripper will be actively maintained at a constant height above the ground using ultrasonic sonar. When a slug is found, it will be picked up by the gripper, which will be already directly over it, and transferred to an on-board storage container. Then, the arm will be returned to its former location so that scanning can continue.

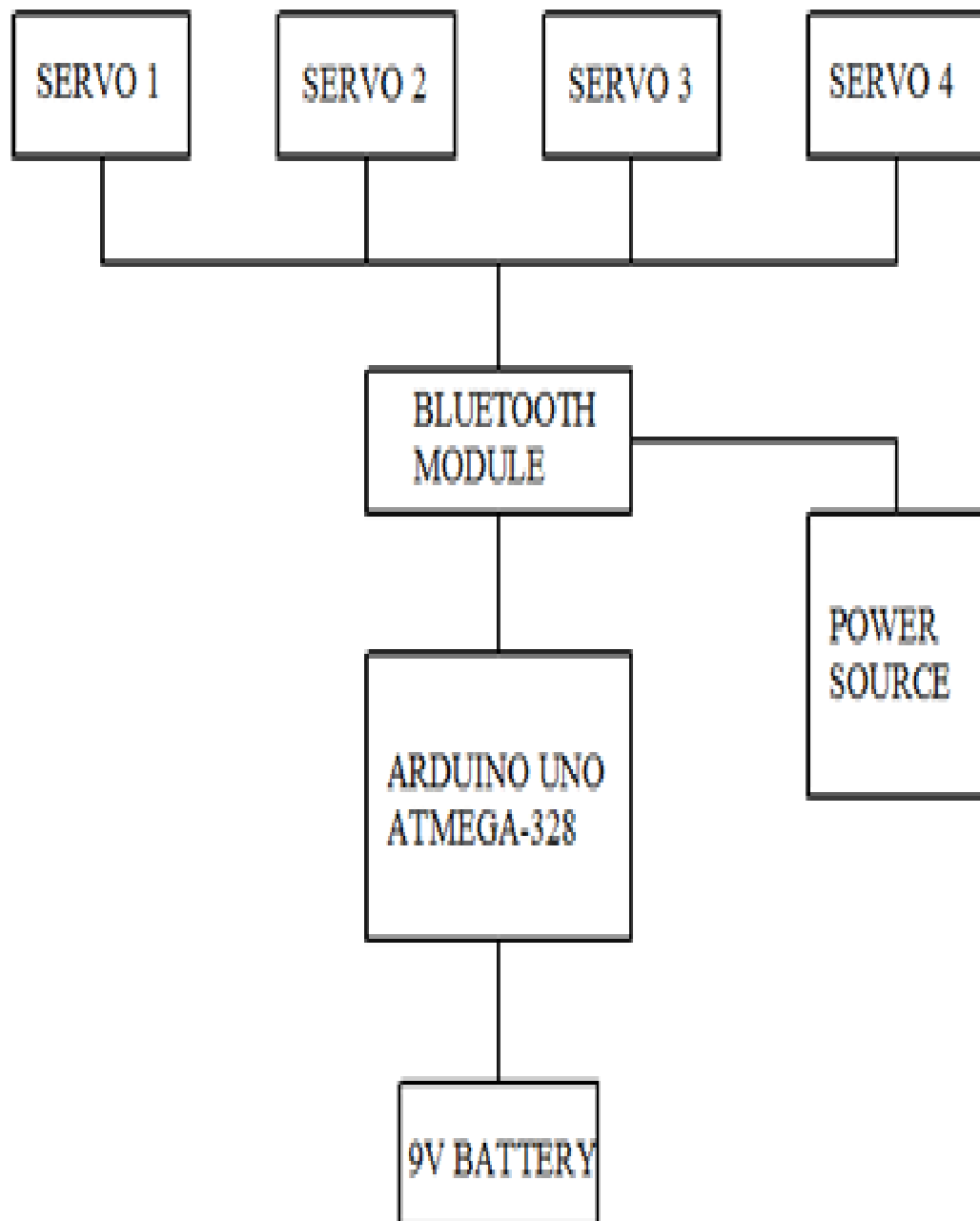
The arm's end creator is a robust lightweight gripper capable of picking up and releasing both wet and dry slugs which regardless of their size, orientation, sliminess, and in the presence of irregularities in the substrate. The version consists of three ngers at 1200 spacing, operated by a single miniature motor shown in figure 2.3. As the ngers close, they meet underneath the slug so that it can be lifted; when the gripper is opened, three wiper blades ensure the slug's release, however much slime is present.

The vision-based system is used to sensor the slug detection in this robot. Since this sensor offers the best combination of size, weight, cost, and effectiveness. VLSI Vision Ltd. Produce monochrome CMOS, single 'chip', image sensor that is small, lightweight, relatively low power (<175m W), of adequate resolution (164 by 124 pixels), and sensitive (down to 0.1 Lux). It is inexpensive, has a digital interface, and the maximum frame rate of 60 Hz enables reasonably high slug scanning speeds. This image sensor also has adjustable automatic exposure control and can calculate the average image intensity of the last frame, and perform pixel level threshold using an adjustable threshold.

The robot system is quite complex. There are a number of motors (gripper, elbow, shoulder, turntable, drive, steering) which must be precisely controlled, and several

sensors (shaft encoders, imaging, obstacle avoidance, scan level, battery level, limit switches etc.) which must be monitored. An added complication is that of the turntable, both power and control signals must be transmitted from the turntable to the main robot base and vice-versa, but many wires would become too twisted and eventually breaks.

1.2 Block diagram



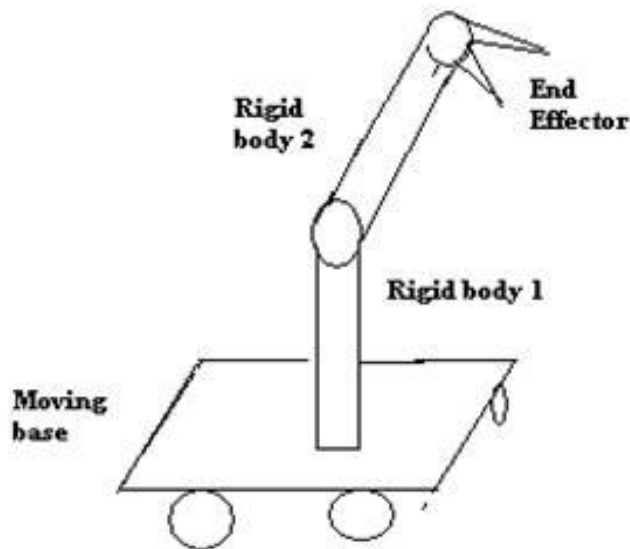


Figure 1.2 robot body

1.2.1 Description

The basic function of a pick and place robot is done by its joints. Joints are analogous to human joints and are used to join the two consecutive rigid bodies in the robot. They can be rotary joint or linear joint. To add a joint to any link of a robot, we need to know about the degrees of freedom and degrees of movement for that body part. Degrees of freedom implement the linear and rotational movement of the body and Degrees of movement imply the number of axis the body can move.

A simple pick and place robot consists of two rigid bodies on a moving base, connected together with rotary joint. A rotary joint is a one which provides rotation in 360 degrees around any one of the axes.

- The bottom or the base is attached with wheels which provide linear movement.
- The 1st rigid body is fixed and supports the second rigid body to which the end effector is provided.
- The 2nd rigid body is provided with movement in all 3 axes and has 3 degrees of freedom. It is connected to the 1st body with a rotational joint.
- The end effector should accommodate all 6 degrees of freedom, in order to reach all sides of the component, to take up position to any height.

On a whole, the basic pick and place robot works as follows:

- The wheels underneath the base help to move the robot to the desired location.
- The rigid body supporting the end effector bends or straightens up to reach the position where the object is placed.
- The end effector picks up the object with a strong grip and places it at the desired position.

Now that we have got a brief idea of the pick and place robot, the basic question is how it is actually controlled.

A simple pick and place robot can be controlled by controlling the movement of its end effector. The motion can be using hydraulic motion, i.e. using hydraulic fluid under pressure to drive the robot, or using pneumatic motion, i.e. using pressurized air to cause mechanical motion. However the most effective way is using motors to provide the required motion. The motors have to be controlled in order to provide required motion to the robot and the end effector.

1.3 Circuit Diagram

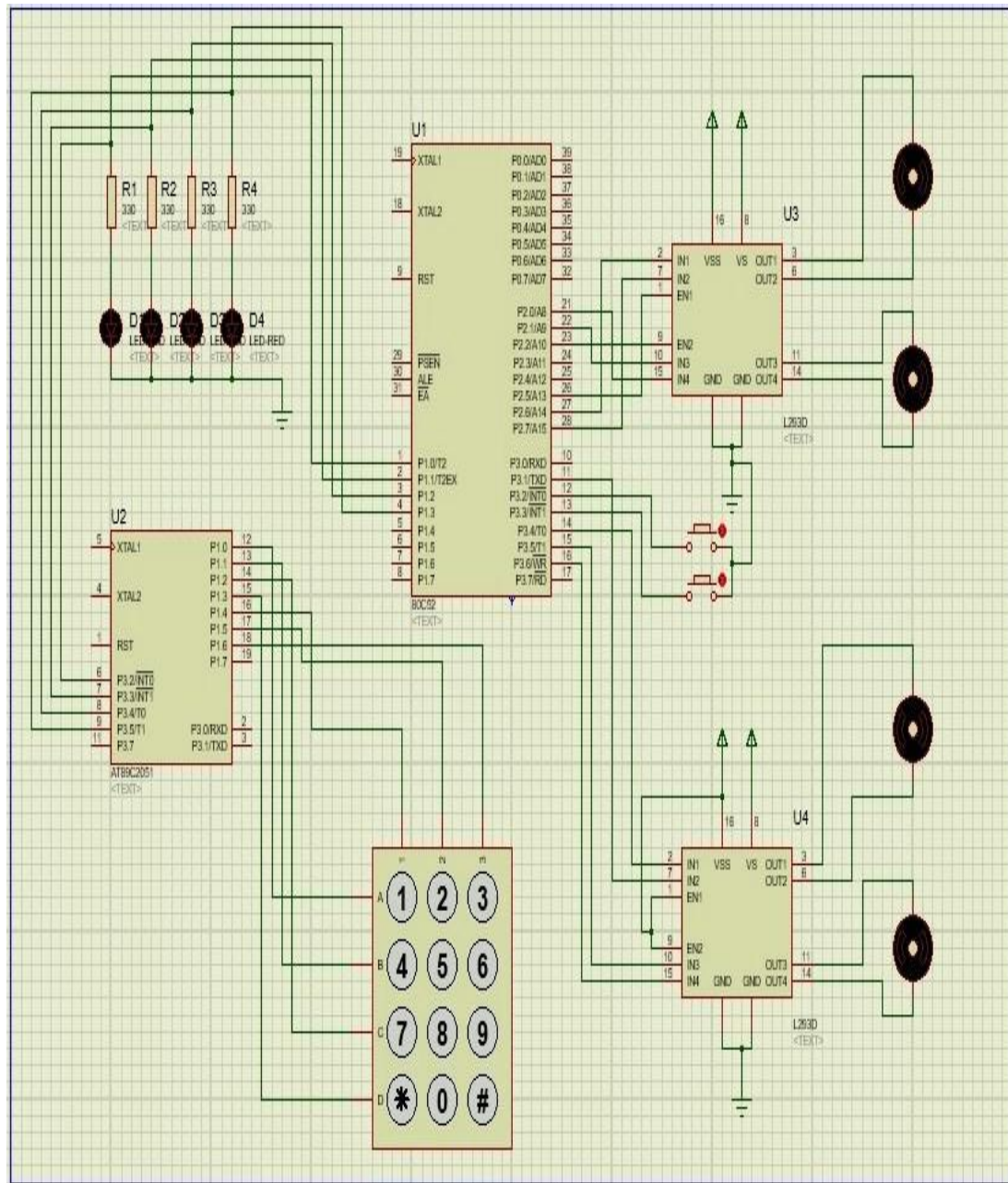


Fig 1.3 Proteus Circuit of Project

1.3.1 Description

There are 4 motors out of which 2 motors are for robotic movement forward and backward, left right and other two motors are independently controlled for certain movement like hand movement or which can be used as a pick and place arrangement. As usual we have a transmitting unit which is comprising of a 20 pin microcontroller AT89C2051/AT89C4051 and a receiving unit comprising of a 40 pin microcontroller 80C51/80C52.

For the purpose of simulation we are connecting 4 data bus of transmitter (AT89C2051) to the receiver microcontroller (80C52). Some LEDs are connected to see data out. On the receiver front two motor driving ICs L293D are connected.

For operating the robot we have taken into effect one matrix keypad. This matrix keypad could be any telephone keypad. This matrix keypad will do all the necessary functions that are required. Also we have got the interrupts pins of the microcontroller 80C52, i.e. INT0 and INT1 used for the purpose of interrupts. Interrupts are required when one of the motor is going into the lock condition. For instance, it may happen that arm is fully open or fully closed. When it is so, the motor goes into locked rotor condition. In that situation interrupt can come and stop the motor. This is the purpose of using the interrupt here.

When we simulate this we find that, when we press 1 the motor-2 rotates in clockwise. When we press 3, motor-1 rotates in clockwise. When pressed 5, both motor-1 and motor-2 rotates anticlockwise. When pressed 2, both of the motors, motor-1 and motor-2 rotates clockwise. The * button is used for the stop function. So we can achieve the following command as far the robotic movement is considered:

Button Pressed	Motor-1	Motor-2	Operation
1	-	Clockwise	Left
3	Clockwise	-	Right
2	Clockwise	Clockwise	Forward
5	Anticlockwise	Anticlockwise	Backward
*	-	-	STOP

Table 1.1 Robotic movement operations

For the other two motors, when 7 is pressed motor-4 rotates in anticlockwise. When 9 is pressed motor-4 rotates in clockwise. This could be used for arm operation, whether the arm is opening or closing. For making the arm up and down, we have 8 button on matrix keypad which will rotate motor-3 in up (clockwise) and 0 for making motor-3 rotate down (anticlockwise). At any moment we can stop it by pressing *

Now suppose the arm up and down movement is going on and an interrupt comes on because the limit is reached, it also stops the motor. Similarly, for the opening and closing functions are carried out with gripper and if the limit is reached, then also interrupt stops the motor. Interrupt can either come from hardware like some photo-sensing device which will force a 0 to these pins when the limit.

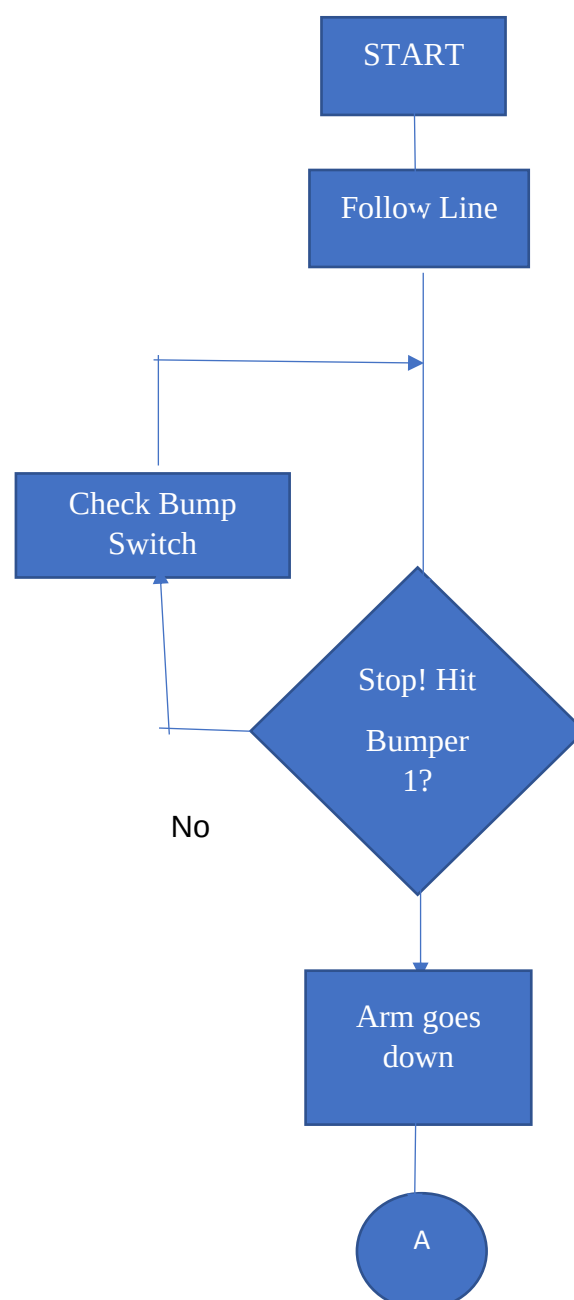
Button Pressed	Motor-3	Motor-4	Operation
7	-	Anticlock wise	Grip Open
9	-	Clockwise	Grip Close
8	Clockwise	-	Arm up
0	Anticlock wise	-	Arm Down
*	-	-	STOP

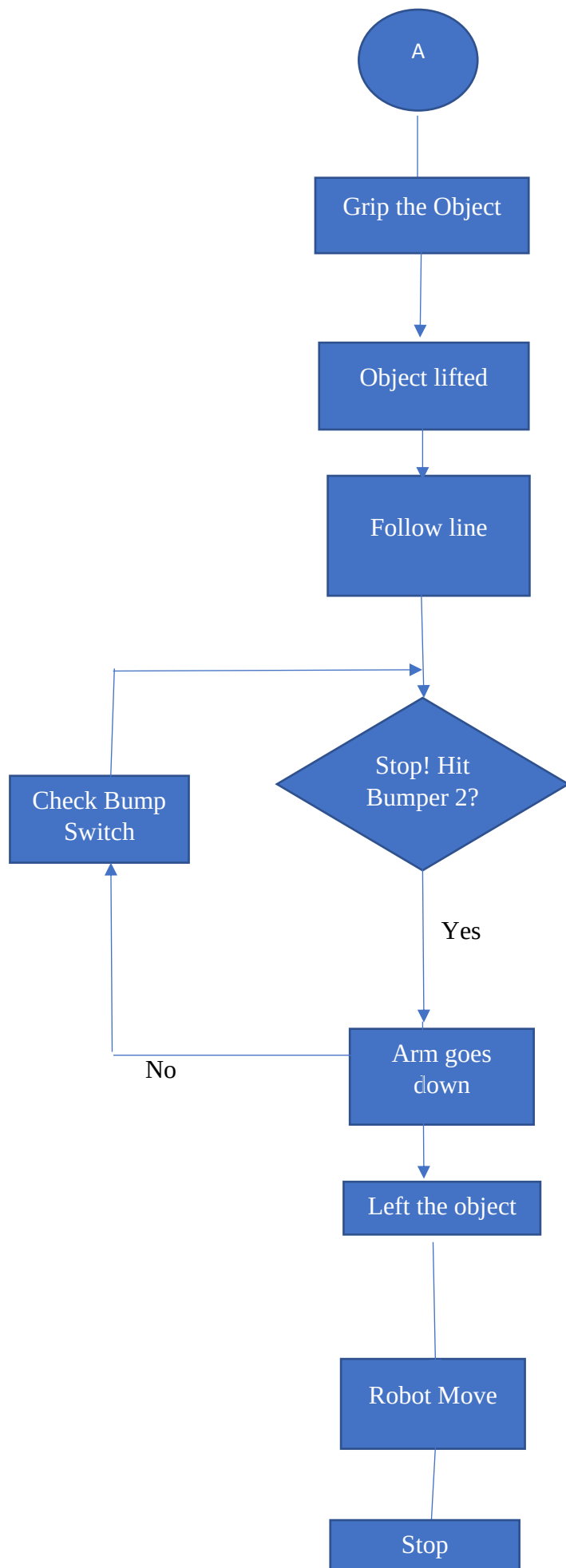
Table 1.2 Arm and Grip movement operations

1.4 Program code

The program is coded using c programming through Arduino software and fed into Arduino board and the program is simulated through proteus software. The power supply is given through adapter and power bank when the board is energized smartphone Bluetooth connects with HC module The customized Application is made specially for robotic arm can operate the robotic arm Three sg servo motors are used for wrist roll, wrist pitch hand gripper.MG servo motors are used waist and shoulder.

1.4.1 Flow charts of the program:





1.4.2 Code for Pick And Place Robot:

```
#include <Servo.h>

Servo servo0;
Servo servo1;
Servo servo2;
Servo servo3;
int sensorvalue0;
int sensorvalue1;
int sensorvalue2;
int sensorvalue3;
void setup()
{
  pinMode(A0,INPUT);
  pinMode(3,OUTPUT);
  servo0.attach(3);

  pinMode(A1,INPUT);
  pinMode(5,OUTPUT);
  servo1.attach(5);

  pinMode(A2,INPUT);
  pinMode(6,OUTPUT);
  servo2.attach(6);

  pinMode(A3,INPUT);
  pinMode(9,OUTPUT);
  servo3.attach(9);
}

void loop()
{
  sensorvalue0 = analogRead(A0);
  sensorvalue0 = map(sensorvalue0, 0, 1023, 0, 180);
  servo0.write(sensorvalue0);
  sensorvalue1 = analogRead(A1);
  sensorvalue1 = map(sensorvalue1, 0, 1023, 0, 180);
  servo1.write(sensorvalue1);
  sensorvalue2 = analogRead(A2);
  sensorvalue2 = map(sensorvalue2, 0, 1023, 0, 180);
```

```
servo2.write(sensorvalue2);  
sensorvalue3 = analogRead(A3);  
sensorvalue3 = map(sensorvalue3, 0, 1023, 0, 180);  
servo3.write(sensorvalue3);  
}
```

1.5 Applications

This pick and place robot project can be used to our advantage in many ways

Defence Applications:

It can be used for surveillance and also to pick up harmful objects like bombs and diffuse them safely.

Industrial Applications: These robots are used in manufacturing, to pick up the required parts and place it in correct position to complete the machinery fixture. It can be also used to place objects on the conveyer belt as well as pick up defective products from the conveyer belt.

Medical Applications: These robots can be used in various surgical operations like in joint replacement operations, orthopaedic and internal surgery operations. It performs the operations with more precision and accuracy.

Other than these applications, these robots can also be used in various other applications suitable to mankind.

Other Advantages:

Before moving further, let us see few reasons why pick and place robots are preferred:

- They are faster and can get the work done in seconds compared to their human counterparts.
- They are flexible and have the appropriate design.
- They are accurate.
- They increase the safety of the working environment and actually never get tired.

CHAPTER 2

2.1 BASIC ARDUINO UNO

2.1.1 Introduction:

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online.

Basic Functions of Arduino Technology

- Digital read pin reads the digital value of the given pin.
- Digital write pin is used to write the digital value of the given pin.
- Pin mode pin is used to set the pin to I/O mode.
- Analog read pin reads and returns the value.
- Analog write pin writes the value of the pin.
- Serial. Begins pin sets the beginning of serial communication by setting the rate of bit.

The most popular board in the Arduino line-up is the Arduino UNO. There are other boards like the Arduino Nano and the Arduino Mega, but UNO has been the go-to board for quick prototyping, Arduino Projects.

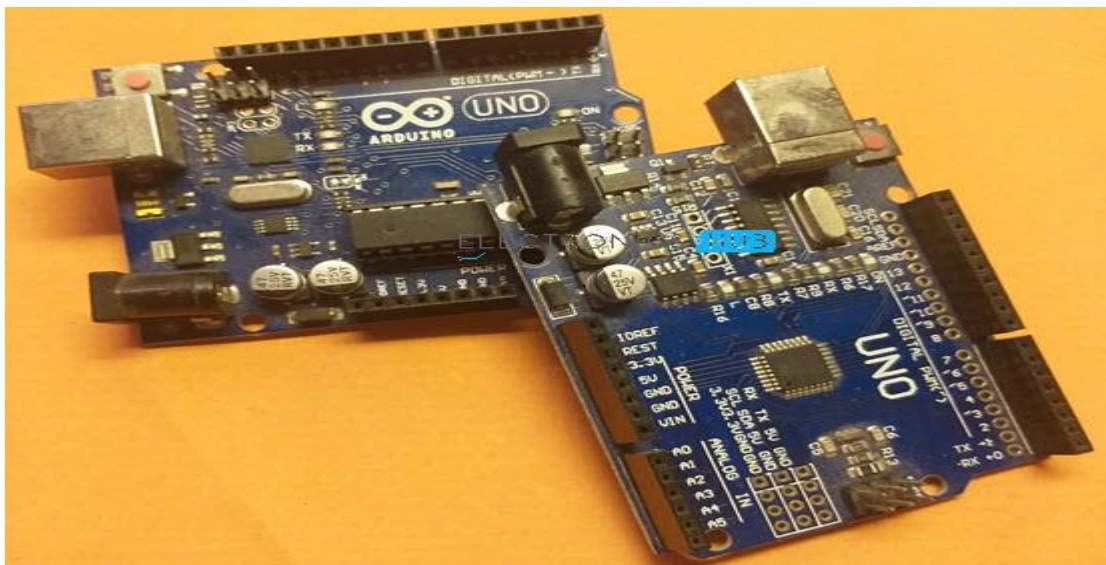


Figure 2.1 Arduino Uno

Arduino UNO is based on ATmega328P Microcontroller, an 8-bit AVR Architecture based MCU from ATMEL. Arduino UNO comes in two variants: one consists of a 28-pin DIP Microcontroller while the other consists of 32 lead Quad Flat Package Microcontroller.

Other than that, rest is identical in both the boards.

2.1.2 Arduino UNO Board Layout:

The following image shows the layout of a typical Arduino UNO board. All the components are placed on the top side of the PCB.

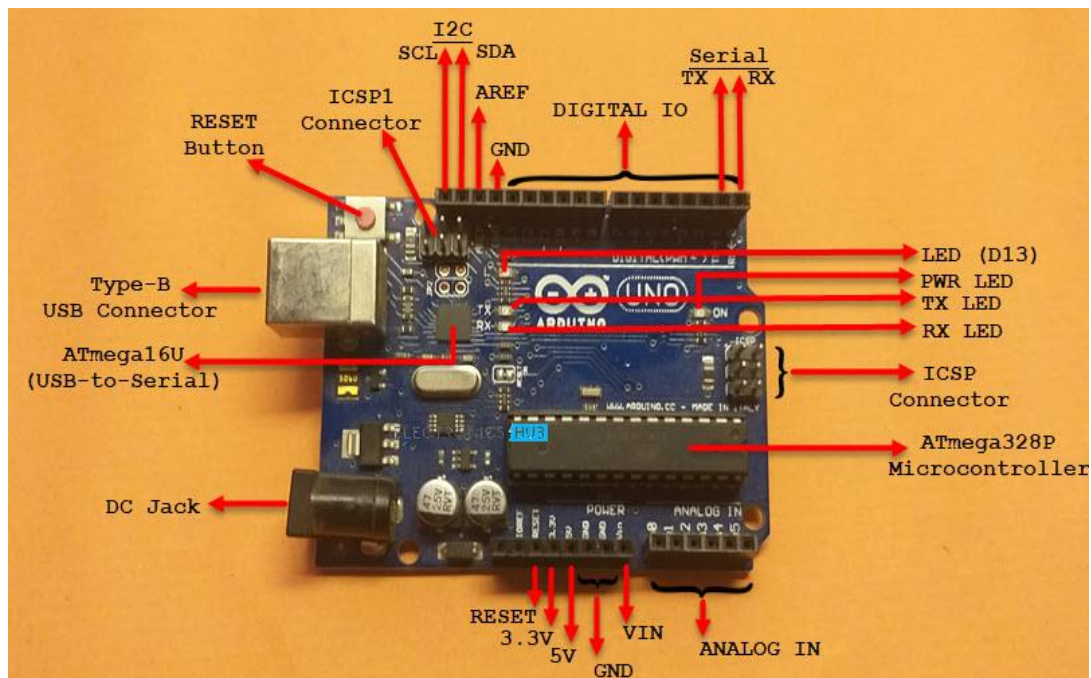


Figure 2.2 Layout of Arduino UNO Board

As you can notice, there is a Type-B USB connector on the left short edge of the board, which is used for powering on the board as well as programming the Microcontroller. There is also a 2.1 mm DC jack to provide external power supply.

Apart from that, the layout of Arduino UNO is very much self-explanatory.

I will discuss about the pins of Arduino UNO in the Arduino UNO Pinout Section.

2.1.3 Memories of Arduino UNO:

Strictly speaking, this is specific to the MCU i.e., ATmega328P, used on the Arduino UNO Board. There are three different memories available in ATmega328P. They are:

- 32 KB of Flash Memory
- 2 KB of SRAM
- 1 KB of EEPROM
- 0.5 KB of the Flash Memory is used by the bootloader code.

2.1.4 Input and Output Pins of Arduino UNO:

Of the 32 pins available on the UNO board, 22 pins are associated with input and output. In that 14 pins (D0 to D13) are true digital IO pins, which can be configured as per your application using `pinMode()`, `digitalWrite()` and `digitalRead()` functions.

All these Digital IO pins are capable of sourcing or sinking 20mA of current (maximum 40mA is allowed). An additional feature of the Digital IO pins is the availability of internal pull-up resistor (which is not connected by default).

The value of the internal pull-up resistor will be in the range of 20K Ω to 50K Ω .

There are also 6 Analog Input Pins (A0 to A5). All the analog input pins provide a 10-bit resolution ADC feature, which can be read using `analogRead()` function.

An important point about Analog Input pins is that they can be configured as Digital IO pins, if required.

Digital IO pins 3, 5, 6, 9, 10 and 11 are capable of producing 8-bit PWM Signals. You can use `analogWrite()` function for this.

2.1.5 Communication Interfaces available on Arduino UNO:

Arduino UNO supports three different types of communication interfaces. They are:

- Serial
- I2C or I²C

- SPI

Perhaps the most common communication interface in the Arduino universe is the Serial Communication. In fact, the Arduino boards (UNO or Nano or Mega) are programmed using the serial communication.

Digital IO pins 0 and 1 are used as Serial RX and TX pins to receive and transmit serial data. These pins are connected to the serial pins of the on-board USB to Serial Converter IC.

Analog Input Pins A4 and A5 have alternative functions. They can be configured as SDA (A4) and SCL (A5) to support I2C or I²C or Two Wire Interface (TWI) communication.

The final communication interface is the SPI. Digital IO Pins 10, 11 12 and 13 can be configured as SPI pins SS, MOSI, MISO and SCK respectively.

2. 1.6 Arduino UNO Pin Layout:

Now that we have seen a little bit about Arduino UNO and its important features and specifications, let us dive into the Arduino UNO Pinout. The following image shows the complete pinout of Arduino UNO Board.

As you can see from the image, I described each pin of the Arduino UNO with its microcontroller equivalent pin, alternative functions, default functionality and other additional features.

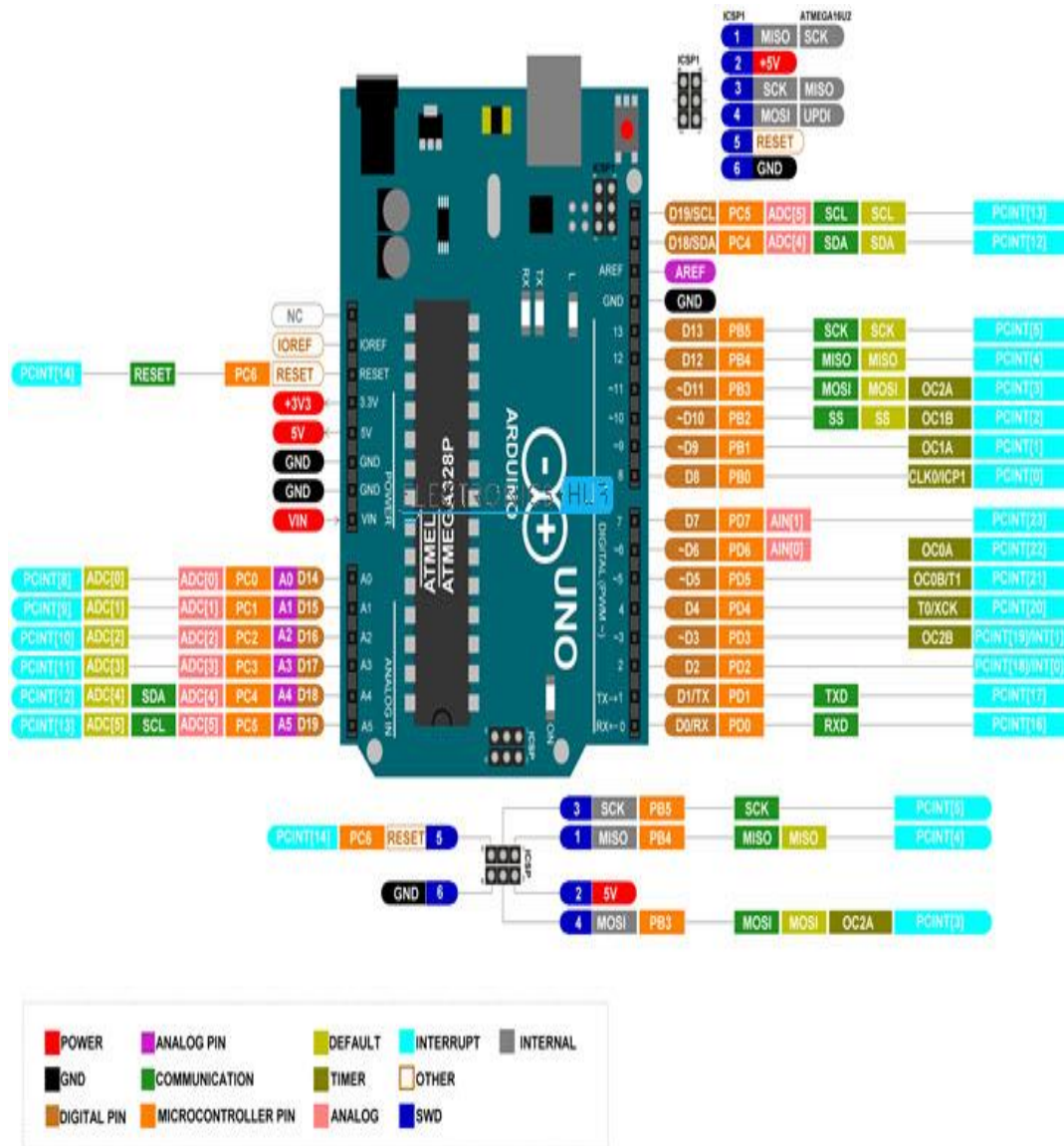


Figure 2.3 Pin Layout

2.1.7 Pin Description:

For pin description of Arduino UNO, let us assume some basic numbering. Let the numbering begin with the RX Pin (D0). So, RX is Pin 1, TX is Pin 2, D2 is Pin 3 and so on. On the other side, NC is Pin 19, IOREF is Pin 20 etc. Overall, there are 32 pins on the Arduino UNO Board.

With this information, let us now see the pin description of Arduino UNO.

Pin Number	Pin Name	Description
1	RX / D0	Digital IO Pin 0 Serial RX Pin
2	TX / D1	Digital IO Pin 1 Serial TX Pin
3	D2	Digital IO Pin 2
4	D3	Digital IO Pin 3
5	D4	Digital IO Pin 4
6	D5	Digital IO Pin 5
7	D6	Digital IO Pin 6
8	D7	Digital IO Pin 7
9	D8	Digital IO Pin 8
10	D9	Digital IO Pin 9
11	D10	Digital IO Pin 10
12	D11	Digital IO Pin 11
13	D12	Digital IO Pin 12
14	D13	Digital IO Pin 13
15	GND	Ground
16	AREF	Analog Reference
17	SDA / D18	Digital IO Pin 18
18	SCL / D19	Digital IO Pin 19
19	NC	Not Connected

20	IOREF	Voltage Reference
21	RESET	Reset (Active LOW)
22	3V3	Power
23	5V	+5V Output from regulator or +5V regulated Input
24	GND	Ground
25	GND	Ground
26	VIN	Unregulated Supply
27	A0	Analog Input 0
28	A1	Analog Input 1
29	A2	Analog Input 2
30	A3	Analog Input 3
31	A4	Analog Input 4
32	A5	Analog Input 5

Table 2.1 Arduino Pin Description

Arduino Architecture

Basically, the processor of the Arduino board uses the Harvard architecture where the program code and program data have separate memory. It consists of two memories such as program memory and data memory. Wherein the data is stored in data memory and the code is stored in the flash program memory. The Atmega328 microcontroller has 32kb of flash memory, 2kb of SRAM 1kb of EPROM and operates with a 16MHz clock speed.

MISO	Master In Slave Out (Input or Output)
5V	Supply
SCK	Clock (from Master to Slave)
MOSI	Master Out Slave In (Input or Output)
RESET	Reset (Active LOW)

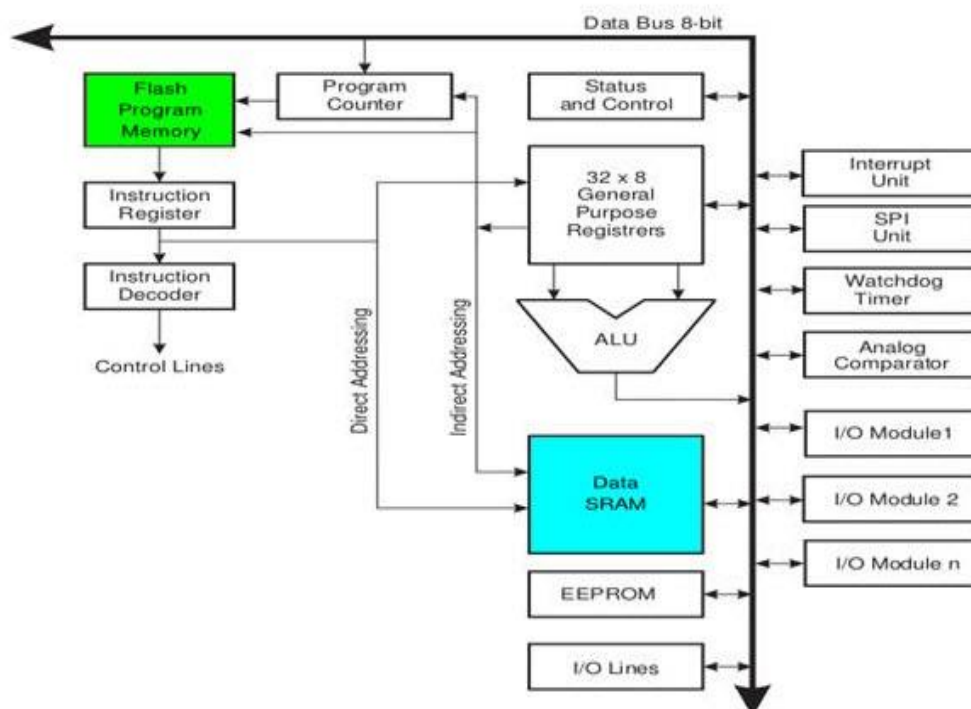
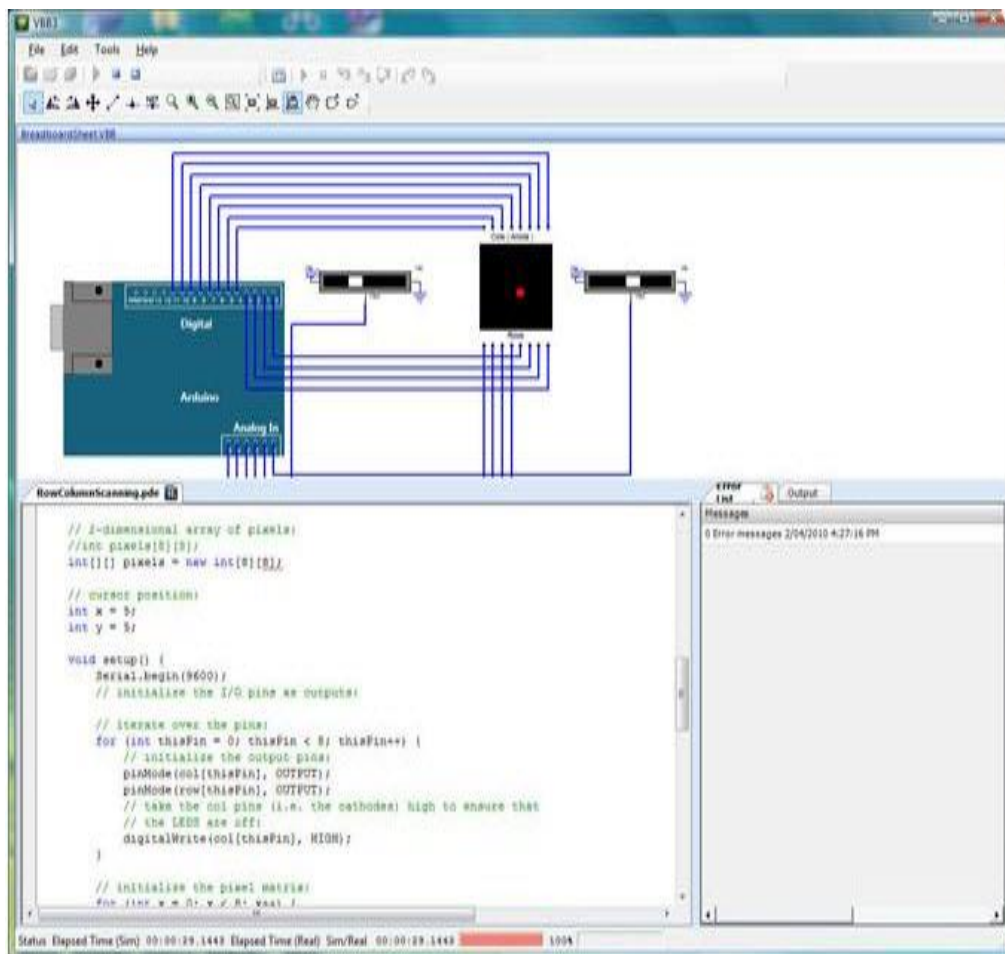


Figure 2.4 Architecture of Arduino

To program an Arduino:

The main advantage of the Arduino technology is, you can directly load the programs into the device without the need of a hardware programmer to burn the program. This is done because of the presence of the 0.5KB of boot loader, that allows the program to be dumped into the circuit. The Arduino tool window contains a toolbar with a various buttons like new, open, verify, upload and serial monitor. And additionally it comprises of a text editor (employed to write the code), a message space (displays the feedback) like showing the errors, the text console, that displays the o/p & a series of menus just like the file, tool menu & edit.



Arduino Programming

- Programming into the Arduino board is called as sketches. Each sketch contains of three parts such as Variables Declaration, Initialization and Control code. Where, Initialization is written in the setup function and Control code is written in the loop function.
- The sketch is saved within and any operation like opening a sketch, verifying and saving can be done using the tool menu.
- The sketch must be stored in the sketchbook directory.
- Select the suitable board from the serial port numbers and tools menu.
- Select the tools menu and click on the upload button, then the boot loader uploads the code on the microcontroller.

CHAPTER 3

Project Components

The design of the Autonomous Pick and Place Mobile Robot which works on ground for the pass section is based on the theory of front wheel design where pulling consume less energy than pushing. Hence, I decided to put the wheel in front after empirical of observation and studies. The concept of the Pick and Place Mobile Robot is based on the AGV (Automated Guided Vehicle) which may look a bit complicated at a glance but it actually consist only 6 main parts. The 7 main parts behind the working of the robot are as follow:

1 HC-05 module

2 RF Transmitter & Receiver

3 Arduino Uno

4 AC Servo Motors

5 Power Supply

6 DIP Switches

7 Programming Pins

All parts mentioned are vital in the functionality of the robot. One cannot work without the others. As such, it is paramount to ensure that each part plays its function properly. Further details about the parts of the project are discussed briefly.

3.1 HC-05 Bluetooth Module

If you take a look around the electronics DIY and hobbyists community, HC-05 Bluetooth Module is the device of choice for implementing Bluetooth Communication based projects. HC-05 Bluetooth Module is a simple Wireless Communication device based on the Bluetooth Protocol.

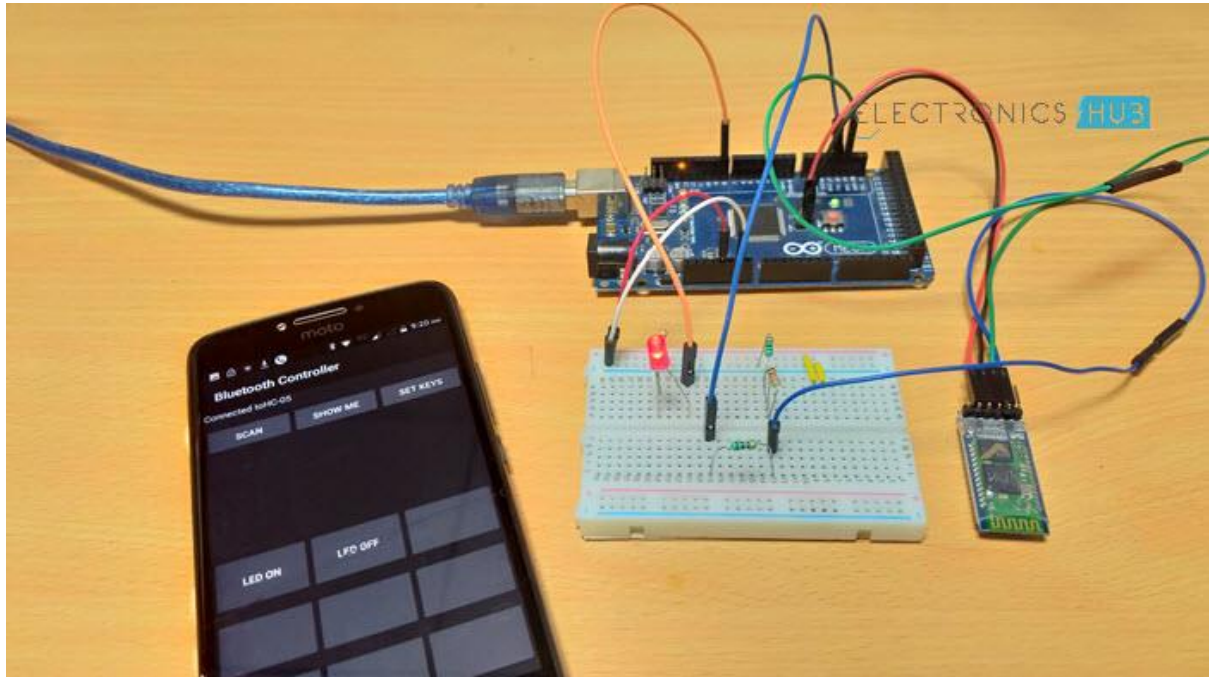


Figure 3.1 HC-05 Bluetooth Module integrating with LED

This module is based on BC417 Single Chip Bluetooth IC that is compliant with Bluetooth v2.0 standard and with support for both UART and USB interfaces.

Generally, the HC-05 Bluetooth Module, or the HC-05 Sub Module, to be precise, comes with the BC417 IC along with a flash memory. Such Modules come as surface mount board and several third-party manufacturers use these board to build a more complete system with necessary pins and components. The following image shows one such HC-05 Bluetooth Sub Module (the green board mounted on the blue board) being used as a part of a complete Bluetooth Module.

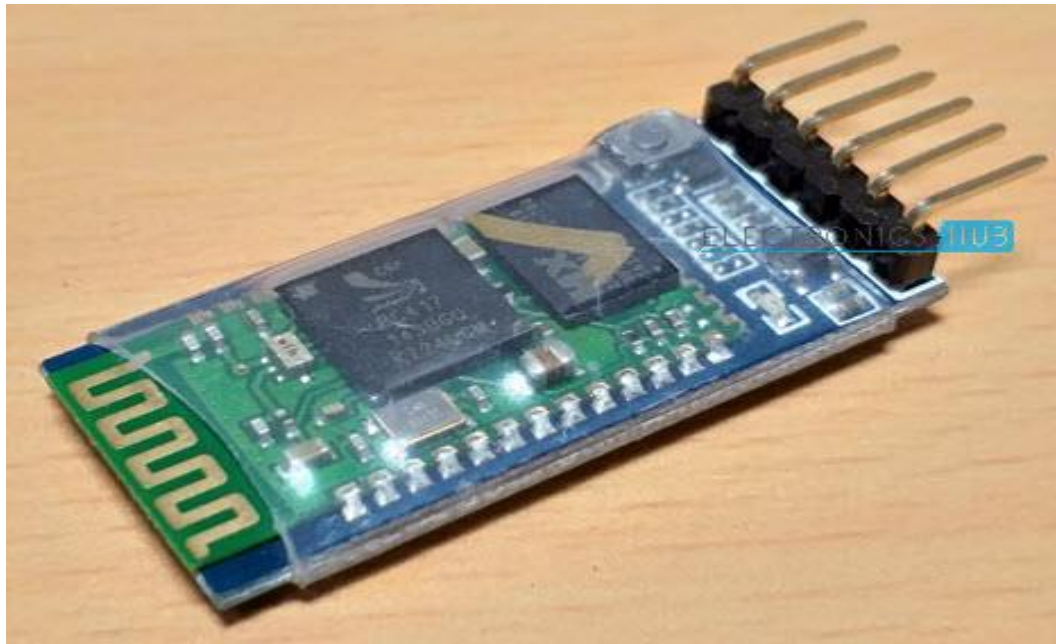


Figure 3.2 HC Module

3.1.1 Pins of HC-05 Bluetooth Module

The HC-05 Module supports for UART, USB as well as SPI communication and depending on the application, necessary pins can be used. In my case, the board uses the UART communication.

Coming to the pins of the Bluetooth Module, generally, four pins are sufficient for successfully enabling a wireless communication link but the modules produced now-a-days come with six pins namely: VCC, GND, TX, RX, EN and STATE.

Image below shows the pins and other components on a typical HC-05 Bluetooth Module.

An important point to remember is the HC-05 Bluetooth Module works on a logic level of 3.3V. Hence, a 3.3V Regulator is used on the board.

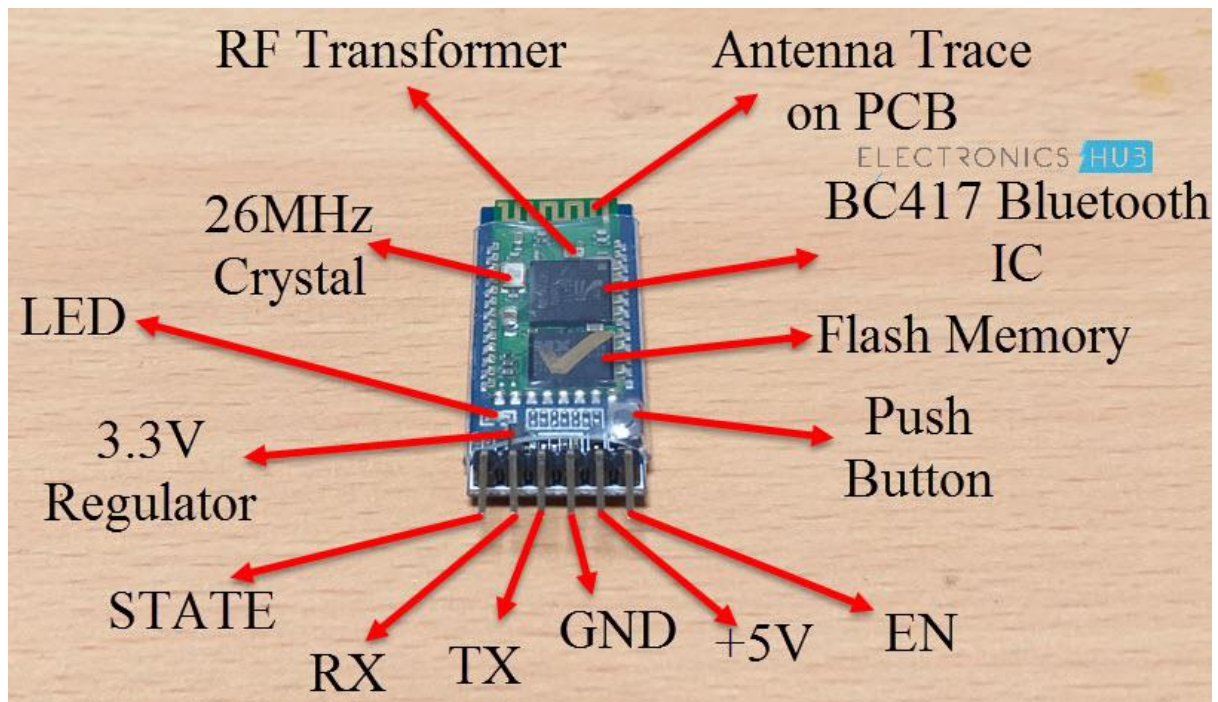


Figure 3.3 HC Module layout

Note: The button present on the board is used to configure the Bluetooth Module in AT Command Mode. This part of the Module is not discussed in this project.

3.1.2 Pin Description

- **EN:** It is the enable pin. When this pin is floating or connected to 3.3V, the module is enabled. If this pin is connected to GND, the module is disabled.
- **+5V:** This is the supply pin for connecting +5V. As the Module has on-board 3.3V regulator, you can provide +5V supply.
- **GND:** It is the ground pin.
- **TX:** It is the Transmitter pin of the UART Communication.
- **RX:** It is the Receive Pin of UART.
- **STATE:** This is a status indicator pin. This pin goes LOW when the module is not connected to any device. When the module is paired with any device, this pin goes HIGH.

Note: The on-board LED is used to indicate the status of the connection. When the module is not paired, the LED blinks or flashes repeatedly. Once the module is paired, the LED blinks at a constant delay of 2 seconds.

3.1.3 Modes of Operation

The HC-05 Bluetooth Module can be configured in two modes of operation: Command Mode and Data Mode.

In Command Mode, you can communicate with the Bluetooth module through AT Commands for configuring various settings and parameters of the Module like get the firmware information, change UART Baud Rate, change module name, set it as either Master or slave etc.

An important point about HC-05 Module is that it can be configured as Master or Slave in a communication pair. In order to select either of the modes, you need to activate the Command Mode and sent appropriate AT Commands.

Coming to the Data Mode, in this mode, the module is used for communicating with other Bluetooth device i.e. data transfer happens in this mode.

3.1.4 Default Settings of HC-05 Bluetooth Module

The following is a list of few of the default settings of the HC-05 Bluetooth Module.

- Name: HC-05
- Password: 1234 (or 0000)
- Type: Slave
- Mode: Data
- Baud Rate: 9600 with 8 data bits, no parity and 1 stop bit

3.1.5 Circuit Diagram for Interfacing HC-05 Bluetooth Module with Arduino

To demonstrate the connection between HC-05 Module and Arduino, I have designed a simple circuit.

The aim of this circuit is to connect the Bluetooth Module with Arduino, Pair the Bluetooth Module with an Android Phone, send data from Android Phone to the Bluetooth Module using a simple App, read the data from Bluetooth Module through Arduino and finally, display the data and control a device based on the data.

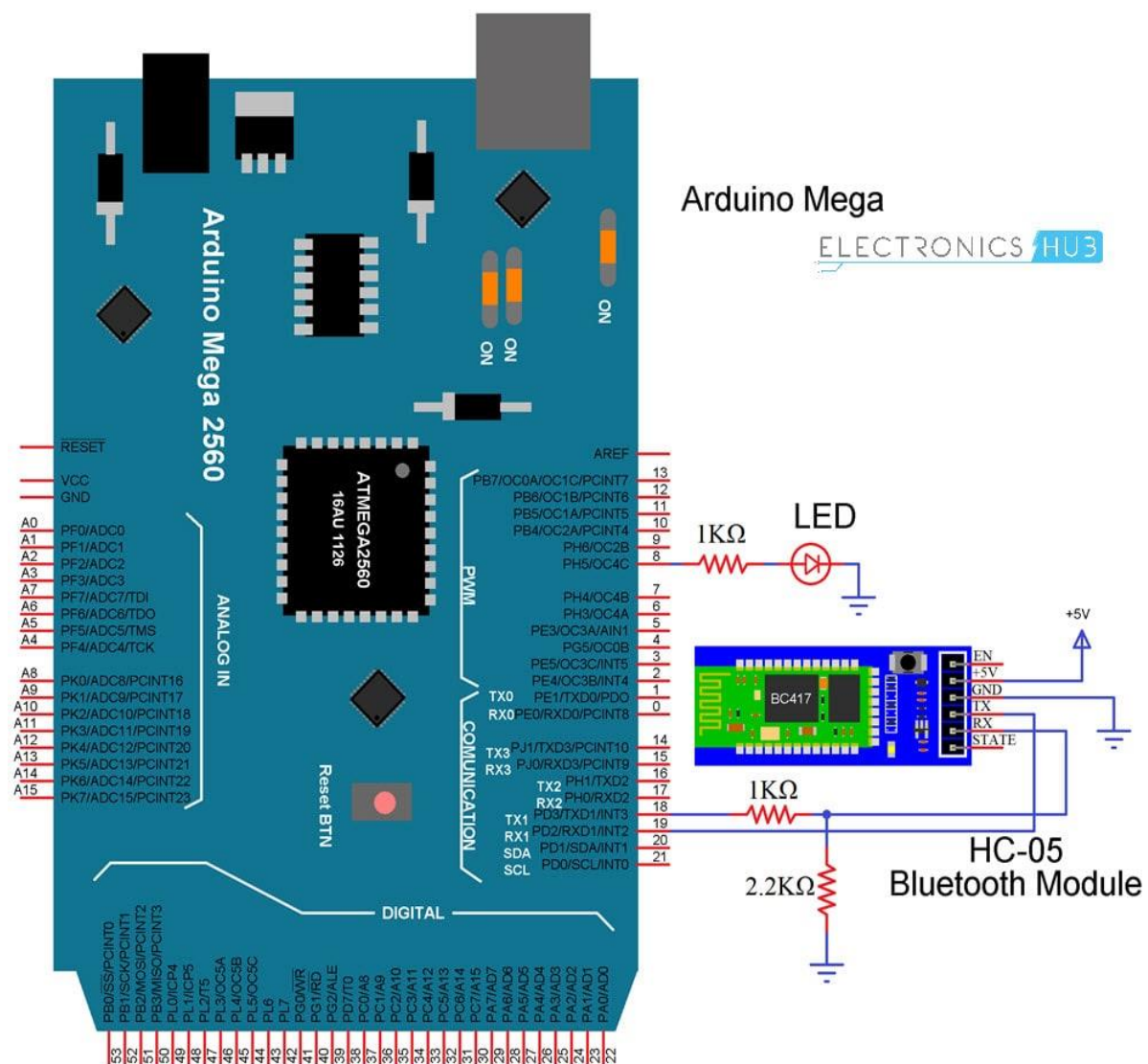


Figure 3.4 Circuit diagram connecting Arduino Mega and HC-05.

Note: I have gone with Arduino Mega just because it has multiple hardware UART Ports. In this project, I am using the default Serial Port of Arduino Mega for communication with the computer (Serial Monitor) and Serial 1 (RX1 and TX1 – Pins 19 and 18) for communication with Bluetooth.

3.1.6 Code

```
/* This code is for Arduino Mega only*/

char data = 0;

void setup()
{
  Serial.begin(9600);
  Serial1.begin(9600);
  pinMode(8, OUTPUT);
}

void loop()
{
  if(Serial1.available() > 0)
  {
    data = Serial1.read();
    if(data == '1')
    {
      Serial.print(data);
      Serial.print("=> LED ON");
      Serial.print("\n");
      /*Serial1.print("LED is ON");
      Serial1.print("\n");*/
      digitalWrite(8, HIGH);
    }
    else if(data == '0')
    {
      Serial.print(data);
      Serial.print("=> LED OFF");
      Serial.print("\n");
      /*Serial1.print("LED is OFF");
      Serial1.print("\n");*/
      digitalWrite(8, LOW);
    }
  }
}
```

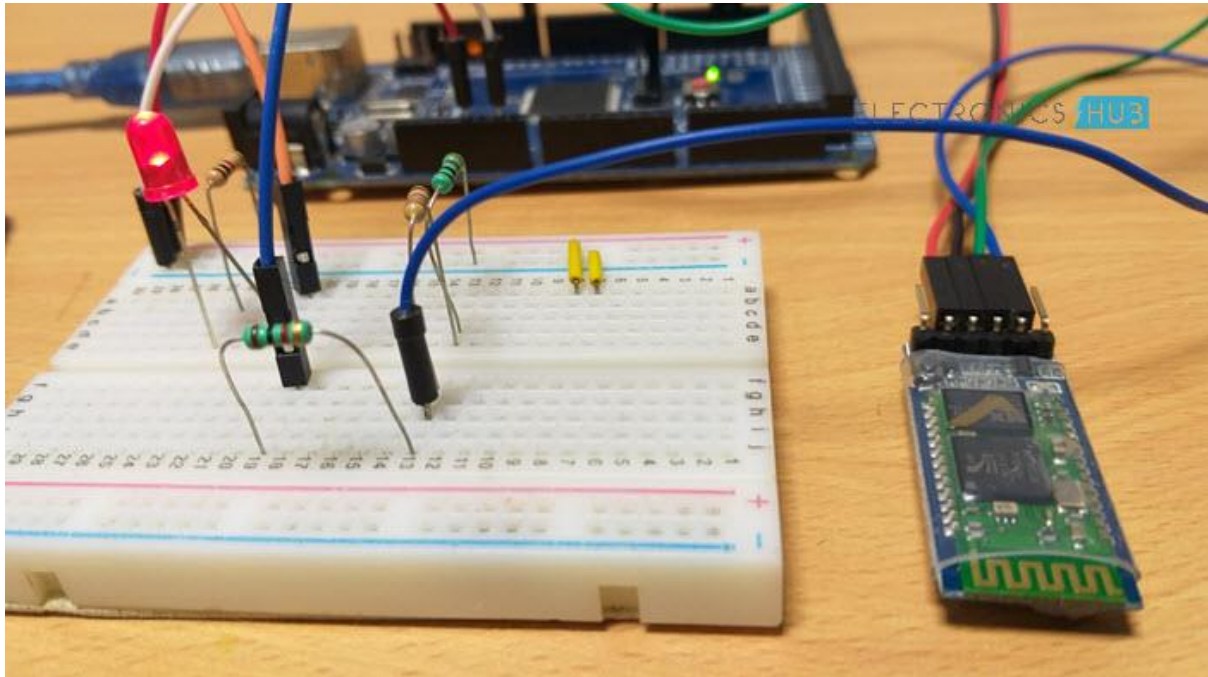


Figure 3.5 HC 05 pin connections

3.1.7 App for Bluetooth Communication

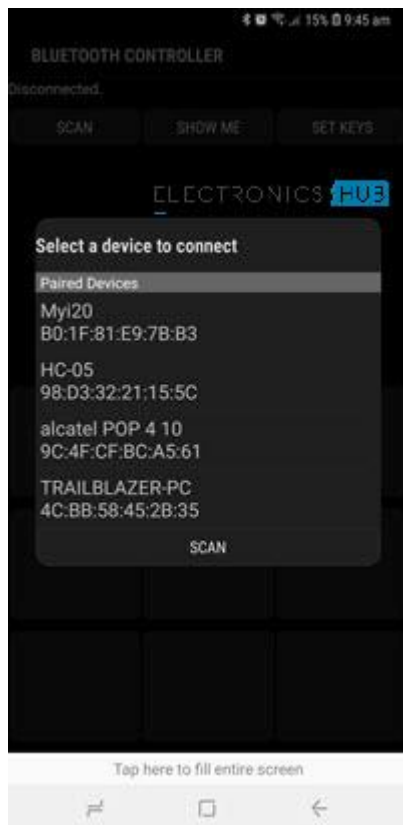
I am using a simple Android App called “Bluetooth Controller” for pairing HC-05 with my Android Phone and transmitting data. At the time of developing this project, this particular Bluetooth Controller App is no longer available in the Play Store.

But I like its simplicity and ease of use. If you want to use the same app, you can download it from the following link **[BLUETOOTH CONTROLLER](#)**.

3.1.8 Connecting Phone to HC-05 Bluetooth Module

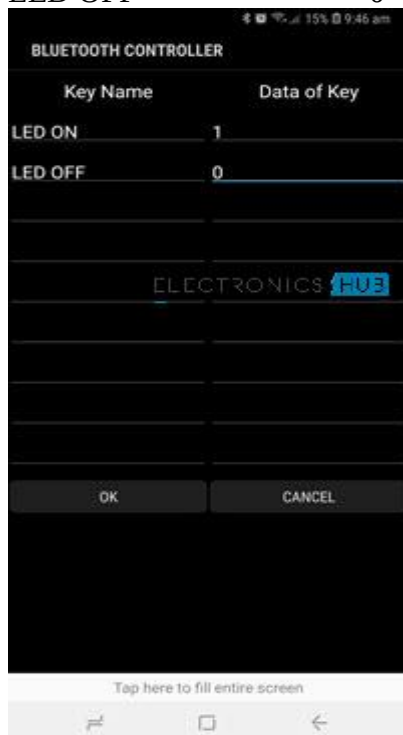
Make the connections and power on the Bluetooth Module. If this is the first time you are using your Bluetooth Module, then the LED will blink rapidly. In order to pair the module with your phone, open Bluetooth Settings in your phone and connect to “HC-05” with pin “1234”. If 1234 doesn’t work, try “0000”.

Once the Bluetooth Module is paired with your phone, you can start using the App. Open the Bluetooth Controller App and click on scan. A list of Bluetooth Devices will appear on the screen. Select HC-05.

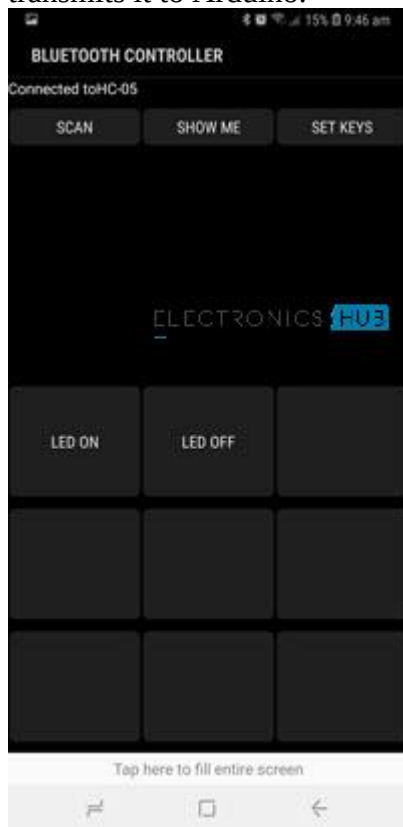


Now, select Set Keys option in the app and enter the following information:

Key Name	Data of Key
LED ON	1
LED OFF	0



After entering this information, save it by clicking on OK. Now, on the home screen of the app, you can see two buttons (or slots) with titles “LED ON” and “LED OFF”. If you click on LED ON, ‘1’ will be transmitted over Bluetooth Communication to the HC-05 Module, which then transmits it to Arduino.



In case you click on LED OFF, ‘0’ will be transmitted. Arduino then turns ON or OFF the LED based on the received data.

3.2 RF Transmitter & Receiver

Radio Frequency: Radio frequency (RF) is a rate of oscillation in the range of around 3 kHz to 300 GHz, which corresponds to the frequency of radio waves, and the alternating currents which carry radio signals.

In this RF system, the digital data is represented as variations in the amplitude of carrier wave. This kind of modulation is known as Amplitude Shift Keying (ASK).

Over 40 millions systems manufactured each year utilizing low-power wireless (RF) technology for data links, telemetry, control and security.

RF Characteristics:

Low power - Typically transmit less than 1mW of power

Good operating range - Operate over distances of 3 to 30 meters Supports data rate up to 1-2 Mbps

Penetrates walls

Does not require a direct transmission path (as opposed to IR)

RF Features:

Serial interface (RS232)

Power supply - 4.5 V dc from three 1.5 V AAA batteries

Operating frequency: 916.50 MHz

Maximum data rate: 22.5 kbps

Operating range: up to 25 meters

{ Obtained in an electrically quiet outdoor location

{ Greatly influenced by building construction materials and contents, other radio systems operating in the vicinity, and noise generated by nearby equipment

Provide link-layer packet protocol

RF Module:

The RF module, as the name suggests, operates at Radio Frequency.

The corresponding frequency range varies between 30 kHz & 300 GHz.

Transmission through RF is better than IR (infrared) Signals through RF

can travel through larger distance.

This RF module comprises of an RF Transmitter and an RF Receiver Tx/Rx pair operates at a frequency of 434 MHz

The transmission occurs at the rate of 1Kbps

Specifications:

Range in open space (standard condition) 100 meter RX receiver

frequency : 433MHz

RX typical sensitivity : 105dBm

Rx supply current : 3.5mA

Rx IF frequency : 1MHz Low power

consumption Easy for application

RX operation voltage : 5V

TX frequency Range : 433.92MHz TX supply

voltage; 3V 6V

TX out put power : 4 12Dbm

Components Used:

IC HT12D,HT12E

TRANSISTOR BC558,SL100

ASK TRANSMITTER 433MHZ ASK

RECEIVER 433MHZ

PUSH TO ON SWITCH RELAY

SWITCH

LED (RED)

RESISTORS

CAPACITORS

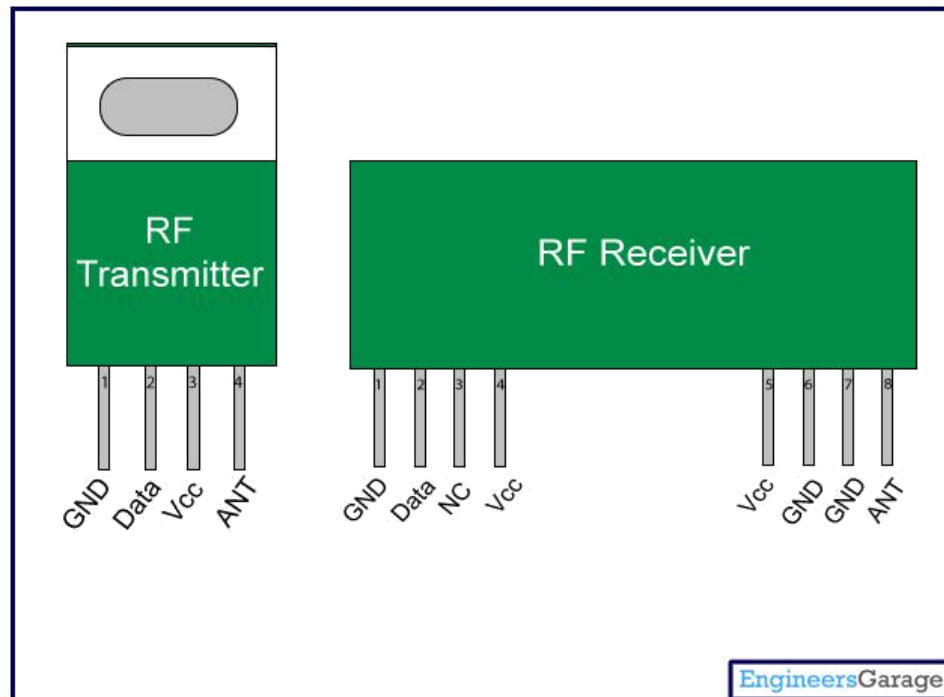


Figure 3.6: Pin Layout

HT12D Decoder:

Features:

Operating voltage: 2.4V 12V

Low power and high noise immunity CMOS technology Low standby current

Capable address setting

Received codes are checked 3 times Address/data

number combination 8 address bits and 4 data bits

HT12E Encoder:

Features:

Operating voltage: 2.4V 12V

Low power and high noise immunity CMOS technology Low standby

current: 0.1A (typ.) at VDD=5V

Four words

Built in oscillator needs only 5 Data code has

positive polarity

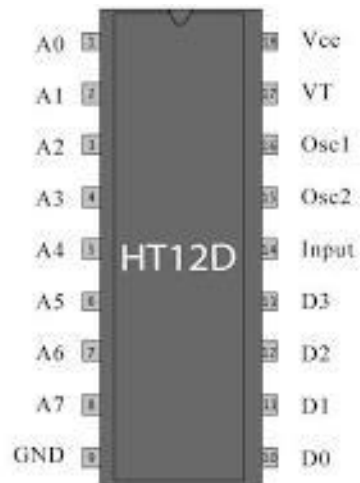


Figure 3.7 Decoder IC HT12D

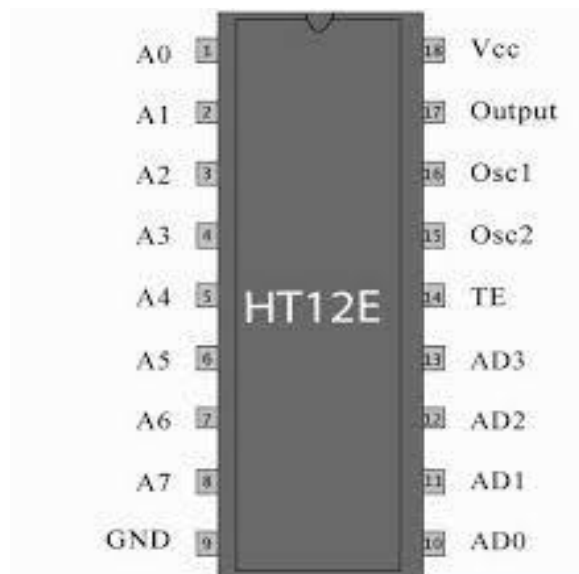


Figure 3.8 Encoder IC HT12E

Pin Number	Description
1-8	A0-A7
9	GND
10-13	AD0-AD3
14	Input
15	OSC 2
16	OSC 1
17	VT
18	Vcc

Table 3.1 Pin Description (HT12D)

Pin Number	Description
1-8	A0-A7
9	GND
10-13	AD0-AD3
14	TE (Active Low)
15	OSC 2
16	OSC 1
17	VT
18	Output

Table 3.2 Pin Description (HT12D)

3.3 Arduino Uno

Arduino UNO R3 board is used in this project. Arduino is a micro controller board based on AT Mega 328p. It has 14 digital input/output pins, 6 analog outputs, 16MHz quartz crystal, a USB connection, a power jack, an ISCP header and a reset button. It powers with a AC-DC adapter or battery to get started. The uno board has 1.0 of Arduino software. HC-05 module is a straightforward to use Blue tooth PP(serial port protocol) designed for clear wireless serial affiliation setup. It can be utilized in Master Slave configuration, making it an excellent resolution for wireless communication. It uses CSR Bluecore04 single chip with CMOS technology and with Adaptive Frequency Hopping.

Of the 32 pins available on the UNO board, 22 pins are associated with input and output. In that 14 pins (D0 to D13) are true digital IO pins, which can be configured as per your application using `pinMode()`, `digitalWrite()` and `digitalRead()` functions.

All these Digital IO pins are capable of sourcing or sinking 20mA of current (maximum 40mA is allowed). An additional feature of the Digital IO pins is the availability of internal pull-up resistor (which is not connected by default).

The value of the internal pull-up resistor will be in the range of 20K Ω to 50K Ω .

There are also 6 Analog Input Pins (A0 to A5). All the analog input pins provide a 10-bit resolution ADC feature, which can be read using `analogRead()` function.

An important point about Analog Input pins is that they can be configured as Digital IO pins, if required.

Digital IO pins 3, 5, 6, 9, 10 and 11 are capable of producing 8-bit PWM Signals. You can use `analogWrite()` function for this.

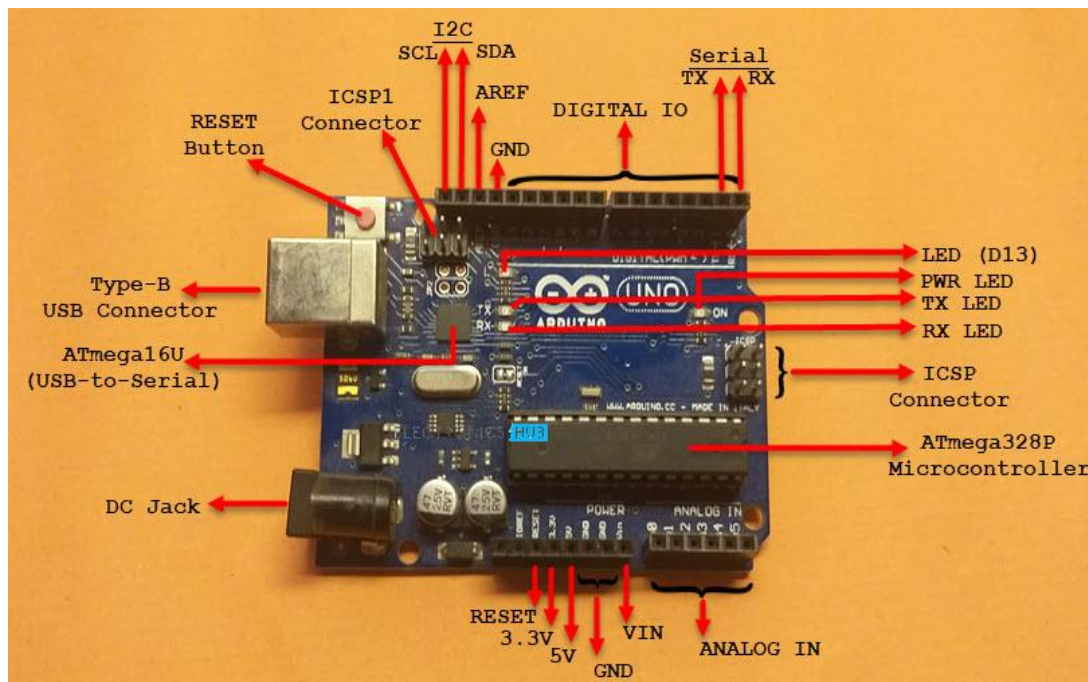


Figure 3.9 Arduino Uno Layout

The processor of the Arduino board uses the Harvard architecture where the program code and program data have separate memory. It consists of two memories such as program memory and data memory. Wherein the data is stored in data memory and the code is stored in the flash program memory. The ATmega328 microcontroller has 32kb of flash memory, 2kb of SRAM 1kb of EPROM and operates with a 16MHz clock speed.

Strictly speaking, this is specific to the MCU i.e., ATmega328P, used on the Arduino UNO Board. There are three different memories available in ATmega328P. They are:

- 32 KB of Flash Memory
- 2 KB of SRAM
- 1 KB of EEPROM
- 0.5 KB of the Flash Memory is used by the bootloader code.

Now that we have seen a little bit about Arduino UNO and its important features and specifications, let us dive into the Arduino UNO Pinout. The following image shows the complete pinout of Arduino UNO Board.

As you can see from the image, I described each pin of the Arduino UNO with its microcontroller equivalent pin, alternative functions, default functionality and other additional features.

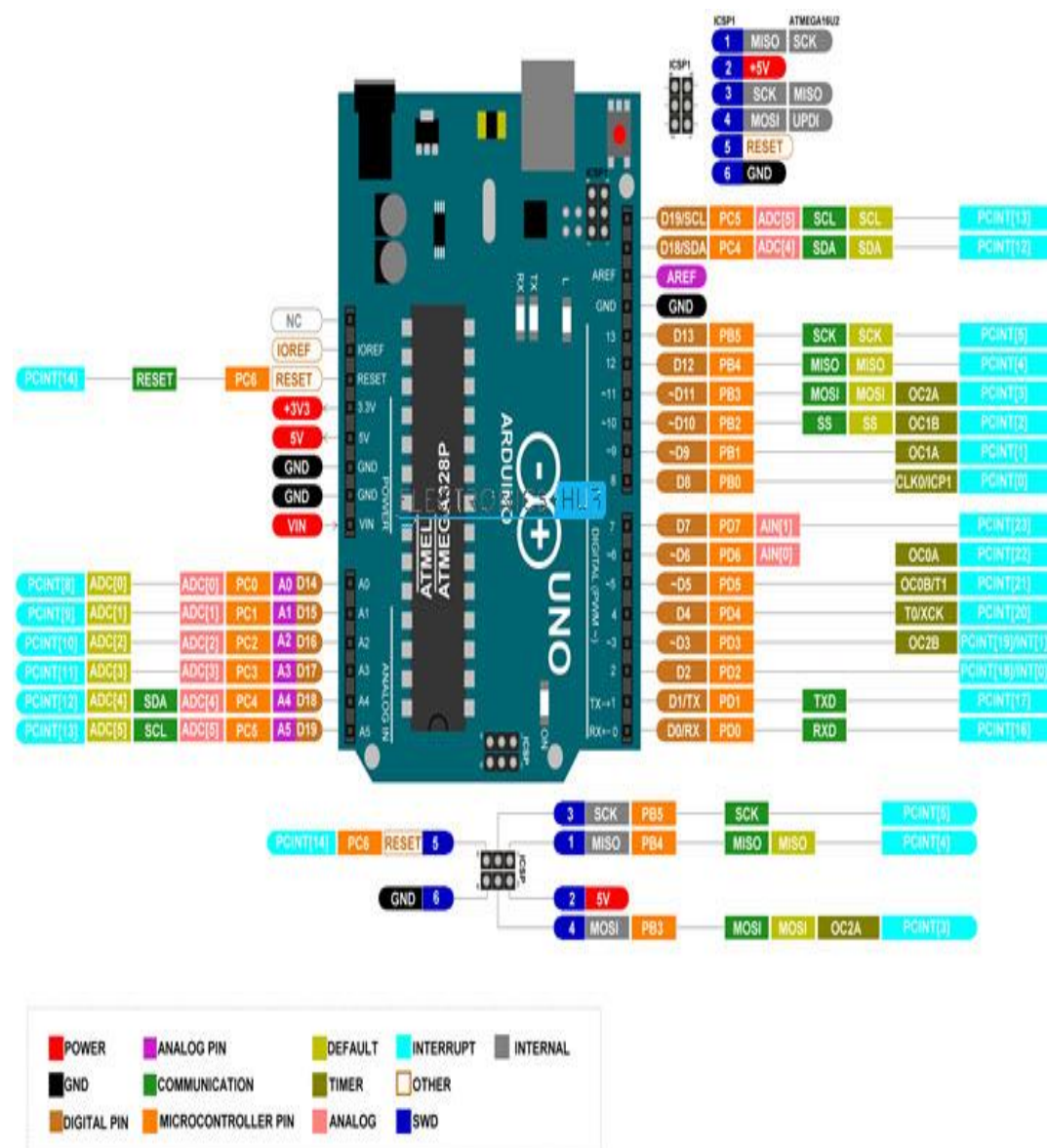


Figure 3.10 Pin Layout

3.4 Motor

Motor is a machine or device that converts any form of energy into mechanical energy, or imparts motion. In constructing a robot, motor plays an important role as to give movement to the robot. In general, motor operates with the effect of conductor with current and the permanent magnetic field. The conductor with current will produce magnetic field which will react with the magnetic field produced by the permanent magnet to make the motor rotate. There are three basic types of motor, DC motor, servo motor and stepper motor which are commonly being used in building a robot.

3.4.1 DC Motor

The DC motor is a device that converts electrical energy into mechanical energy. The DC motor has a rotating armature in the form of an electromagnet. A rotary switch called a commutator reverses the direction of the electric current twice every cycle, to flow through the armature so that the poles of the electromagnet push and pull against the permanent magnets on the outside of the motor. As the poles of the armature electromagnet pass the poles of the permanent magnets, the commutator reverses the polarity of the armature electromagnet. During that instant of switching polarity, inertia keeps the classical motor going in the proper direction.



Figure 3.11 A simple DC Motor

When the coil is powered, a magnetic field is generated around the armature. The left side of the armature is pushed away from the left magnet and drawn toward the right, causing rotation. The armature continues to rotate. When the armature becomes horizontally aligned, the commutator reverses the direction of current through the coil, reversing the magnetic field. The process then repeats.

Current Use:

Used in industrial applications because the speed-torque relationship can be varied to almost any useful form.

Often applied where they momentarily deliver three or more times their rated torque.

Used on applications requiring quick stops.

Features:

Speed can be controlled smoothly down to zero.

Immediately followed by acceleration in the opposite direction without power circuit switching.

Respond quickly to changes in control signals due to the dc motor's high ratio of torque to inertia.

Advantages:

Simple to operate and do not rely upon a control circuit.

Simply attach a DC power supply to the DC motor and it will run. Have stators and rotors.

The rotor motion does not follow stators that are sequentially magnetized.

Disadvantages:

Do not have inherent positional control.

Optical sensors may malfunction because of dust or wear particles created by the mechanical braking system.

Have a negative effect on achieving the accuracy and precision of piston movement required in an application.

Must incorporate additional mechanisms, such as gearing, clutches, and a solenoid brake to control speed and stop the rotor in desired positions.

The reliability of the DC motor system is frequently an issue.

3.4.2 Servo Motor

Servomotors are basically geared down dc motors with positional feedback control, allowing for accurate positioning of the rotor, with a range of 90 degrees. They can also be modified to allow for continuous rotation. The servo is controlled by three wires: ground (black), power (red), and command (typically white). Power is usually between 4 v and 6v and should be separate from system power (as servos are electrically noisy). Servos may be driven to higher voltages to improve torque and speed characteristics. The servo motor has some control circuits and a potentiometer (a variable resistor) that is connected to the output shaft. The pot can be seen on the right side of the circuit board. This pot allows the control circuitry to monitor the current angle of the servo motor. If the shaft is at the correct angle, then the motor shuts o . If the circuit sends that the angle is not correct, it will turn the motor the correct direction until the angle is correct.



Figure 3.12 Servo Motor

Once the servo has received the desired position (via the PWM signal) the servo must attempt to match the desired and actual positions. It does this by turning a small, geared motor left or right. If, for example, the desired position is less than the actual position, the servo will tum to the left. On the other hand, if the desired position is greater than the actual position, the servo will tum to

the right. In this manner, the servo "zeros-in" on the correct position. Should a load force the servo horn to the right or left, the servo will attempt to compensate.

Servos are commanded through "Pulse Width Modulation," or PWM, signals sent through the command wire. This control signal is a variable-width pulse, which can be varied from 1 to 2ms. The pulse width controls the rotor position. A 1.0ms pulse rotates the shaft all the way counter-clockwise. A 1.5ms pulse puts the rotor at neutral (0 degrees), and a 2.0ms pulse will position the shaft all the way clockwise. The pulse is sent to the servo at a frequency of approximately 50 Hz.

Current Use:

Used in Radio Controlled models like cars, airplanes, robots, and puppets. Used in powerful heavy-duty sail boats.

Extremely useful in robotics.

Normally a servo is used to control an angular motion of between 0 and 180.

Features:

Incorporates a three wire DC Motor, a gear train, a potentiometer, an integrated circuit

The shaft of the servo can be positioned to specific angular positions by sending a coded signal.

Draws power proportional to the mechanical load.

Lightly loaded servo, therefore, doesn't consume much energy.

When the control circuit detects that the position is correct, it stops the motor and if detect the incorrect angle, it will turn the motor to the correct direction until find the correct angle.

Advantages:

If a heavy load is placed on the motor, the driver will increase the current to the motor coil as it attempts to rotate the motor. Basically, there is no out-of-step condition. However, too heavy a load may cause an error.

High-speed operation is possible.

Disadvantages:

Since the servomotor tries to rotate according to the command pulses, but lags behind, it is not suitable for precision control of rotation.

Higher cost.

When stopped, the motor's rotor continues to move back and forth one pulse, so that it is not suitable if you need to prevent vibration.

3.5 DIP Switch

DIP Switches are manual electric switches that are packaged by group into a standard dual in-line package (DIP). This type of switch is designed to be used on a printed circuit board along with other electronic components to customize the behavior of an electronic device in specific situations. DIP switches are also known as toggle switches, which mean they have two possible positions { on or o . (Sometimes instead of on and o , you may see the numbers 1 and 0 on the DIP Switch.)

Types:

Slide and Rocker Actuator DIP Switches: These are typical on/o switches with a SPST (single-pole, single-throw) contacts. They have a one-bit binary value with a standard ASCII character.

Rotary DIP Switch: This DIP switch has several electrical contacts which are rotated and aligned. They switches can be small or large and provide a selection of switching combinations.

SPDT (double pole single throw) DPST (double pole single throw) DPDT (double pole double throw) MPST (multiple-pole single-throw) and MTSP (multiple-throw single-pole) DIP switches

Applications: Typical Dip Switch applications include:

Telecommunications

Computer Motherboards

Remote controls for frequency setting (like garage door openers) and other electronic devices requiring option settings

3.6 Power Supply

The robot has 2 different power supply needs. The 5V power supply is vital for the microcontroller, IR sensors and bump switches. The 6V power supply is needed by the 5 servo-motors as the 5V supply is insufficient to run the servo-motors smoothly. That is why 6V (1.3AH) fuel sealed rechargeable battery are used as supply to the Pick and Place Mobile Robot. The purpose of the switch is to turn ON/OFF the robot as a whole including the 2 wheel servomotors. When the switch is turned ON, the 6V supply will go into the LM7806 regulator through the RST pin of the regulator. The output of the regulator from the third pin is a constant 5V supply that is important to the microcontroller, IR sensor and the bump switches. The purpose of the capacitor is to ensure that the supply is constant without irregularities and distortion.

3.7 Programming Pins

There are 4 programming pins for the socket of the programming cable to connect to. This is how the source code of a program is transferred from the computer to the EEPROM of the microcontroller. The programming pins correspond to the first 4 pins of the BS2 microcontroller. One end of the programming cable has a female socket that is designed to be able to connect to the 4 pins of the programming pins. The other end is male socket that is connected to the serial port of the computer.

CHAPTER 4

4.1 Implementation

This project is implemented using following software's:

- ❖ Proteus simulation– for designing circuit
- ❖ Arduino compiler - for compilation part

4.2 About Proteus:

It is a software suite containing schematic, simulation as well as PCB designing.

- **ISIS** is the software used to draw schematics and simulate the circuits in real time .The simulation allows human access during run time, thus providing real time simulation.
- ARES is used for PCB designing .It has the feature of viewing output in 3D view of the designed PCB along with components.
- The designer can also develop 2D drawings for the product.

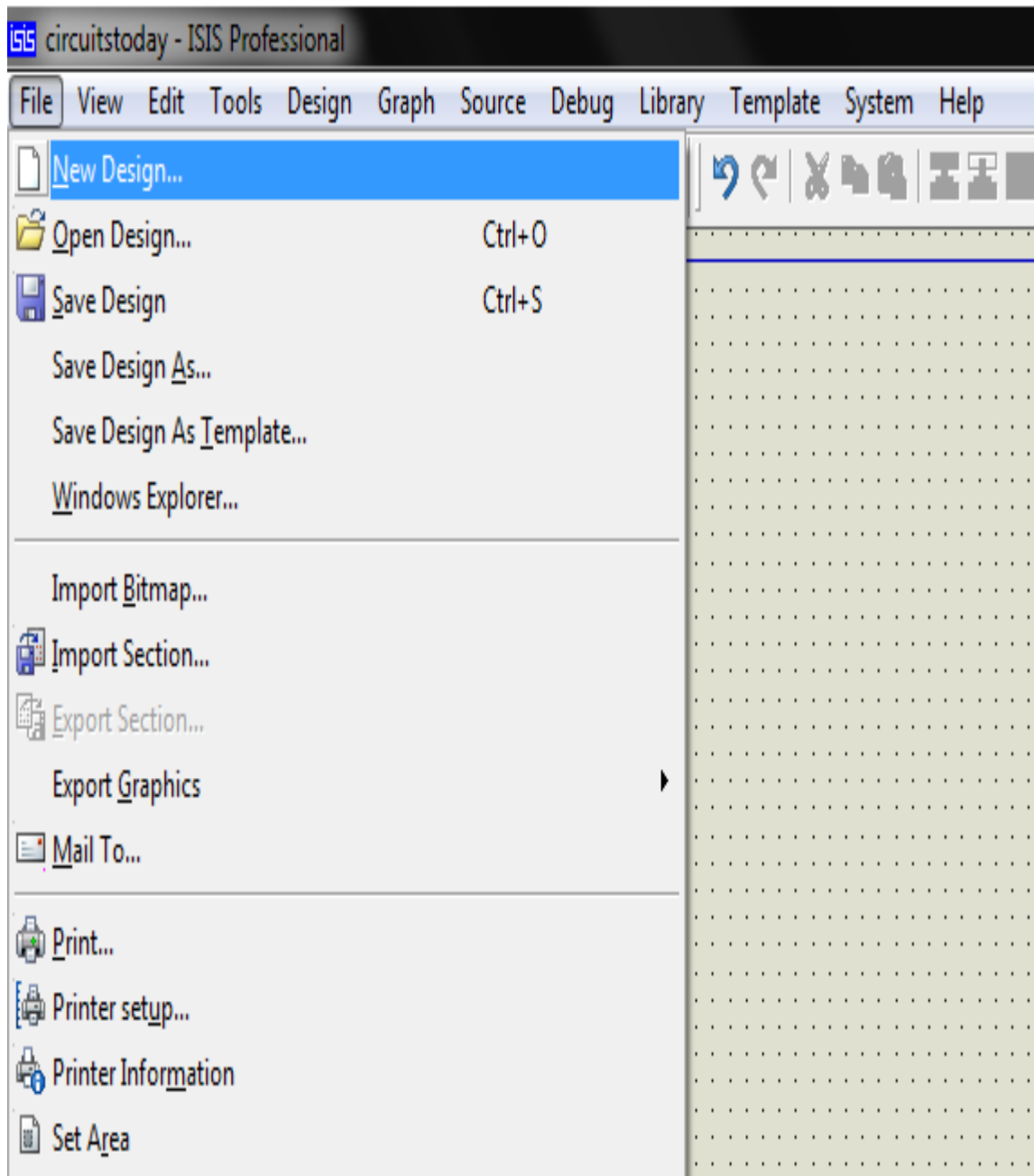
Features:

ISIS has wide range of components in its library. It has sources, signal generators, measurement and analysis tools like **oscilloscope**, voltmeter, ammeter etc., probes for real time monitoring of the parameters of the circuit, switches, displays, loads like motors and lamps, discrete components like resistors, capacitors, inductors, transformers, digital and analog Integrated circuits, semi-conductor switches, relays, microcontrollers, processors, sensors etc.

ARES offers PCB designing up to 14 inner layers, with surface mount and through hole packages. It is embedded with the foot prints of different category of components like ICs, transistors, headers, connectors and other discrete components. It offers Auto routing and manual routing options to the PCB Designer. The schematic drawn in the ISIS can be directly transferred ARES.

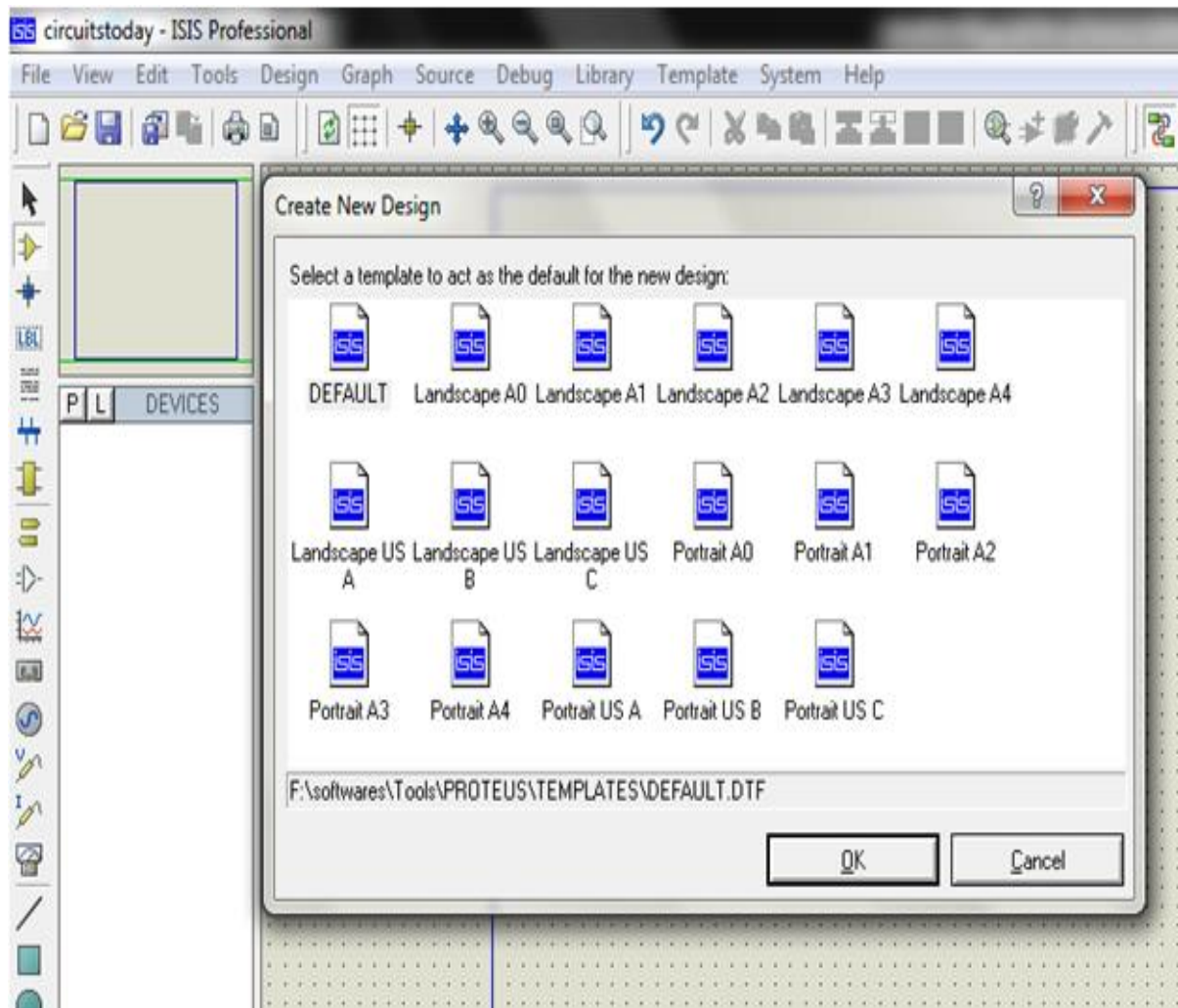
4.3 Starting New Design

Step 1: Open ISIS software and select New design in File menu



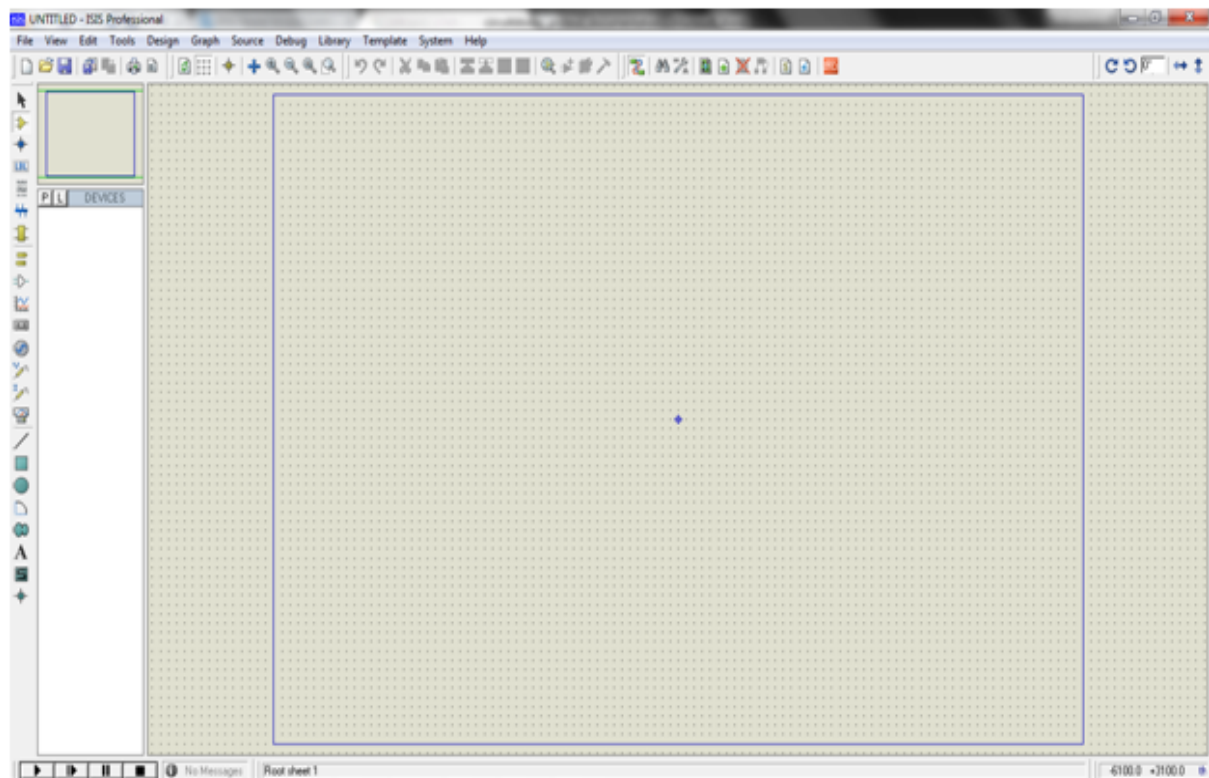
Proteus File Menu

Step 2: A dialogue box appears to save the current design. However, we are creating a new design file so you can click Yes or No depending on the content of the present file. Then a Pop-Up appears asking to select the template. It is similar to selecting the paper size while printing. For now select default or according to the layout size of the circuit.



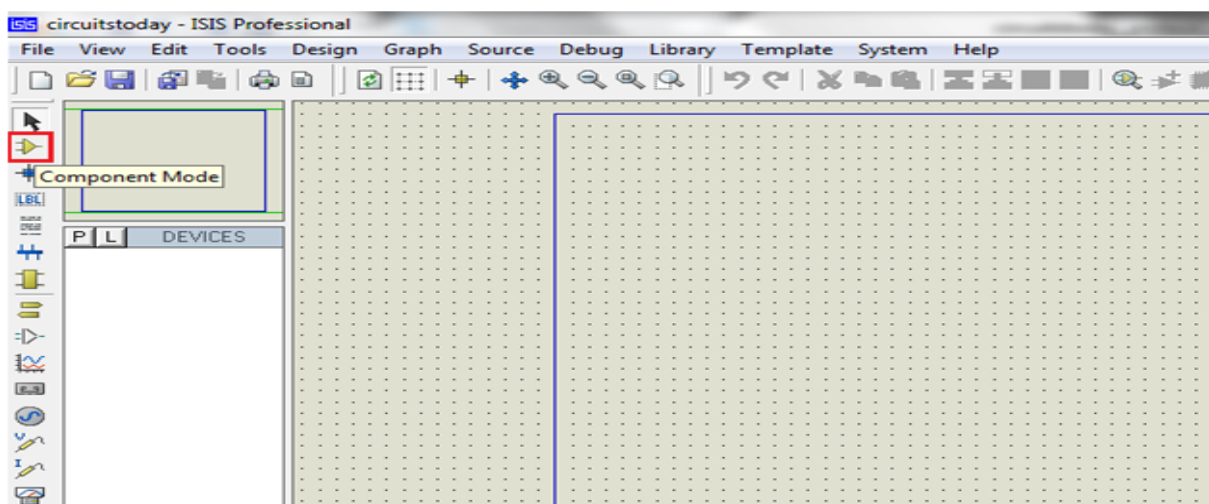
Proteus Default Template Select

Step 3: An untitled design sheet will be opened, save it according to your wish, it is better to create a new folder for every layout as it generates other files supporting your design. However, it is not mandatory.



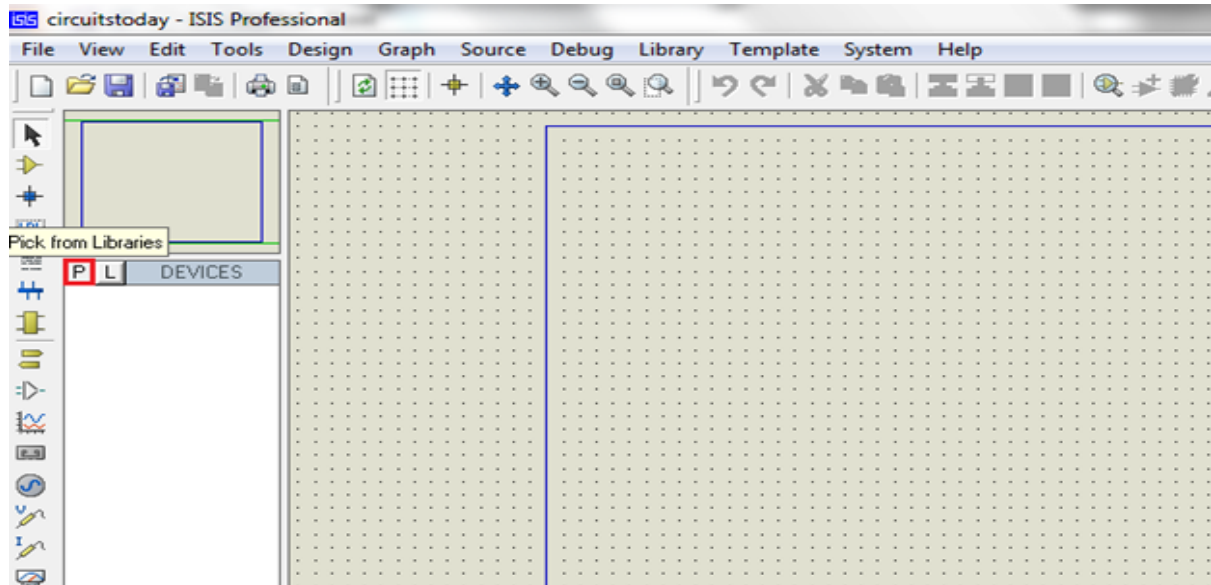
Proteus Design Sheet

Step 4: To Select components, Click on the component mode button.



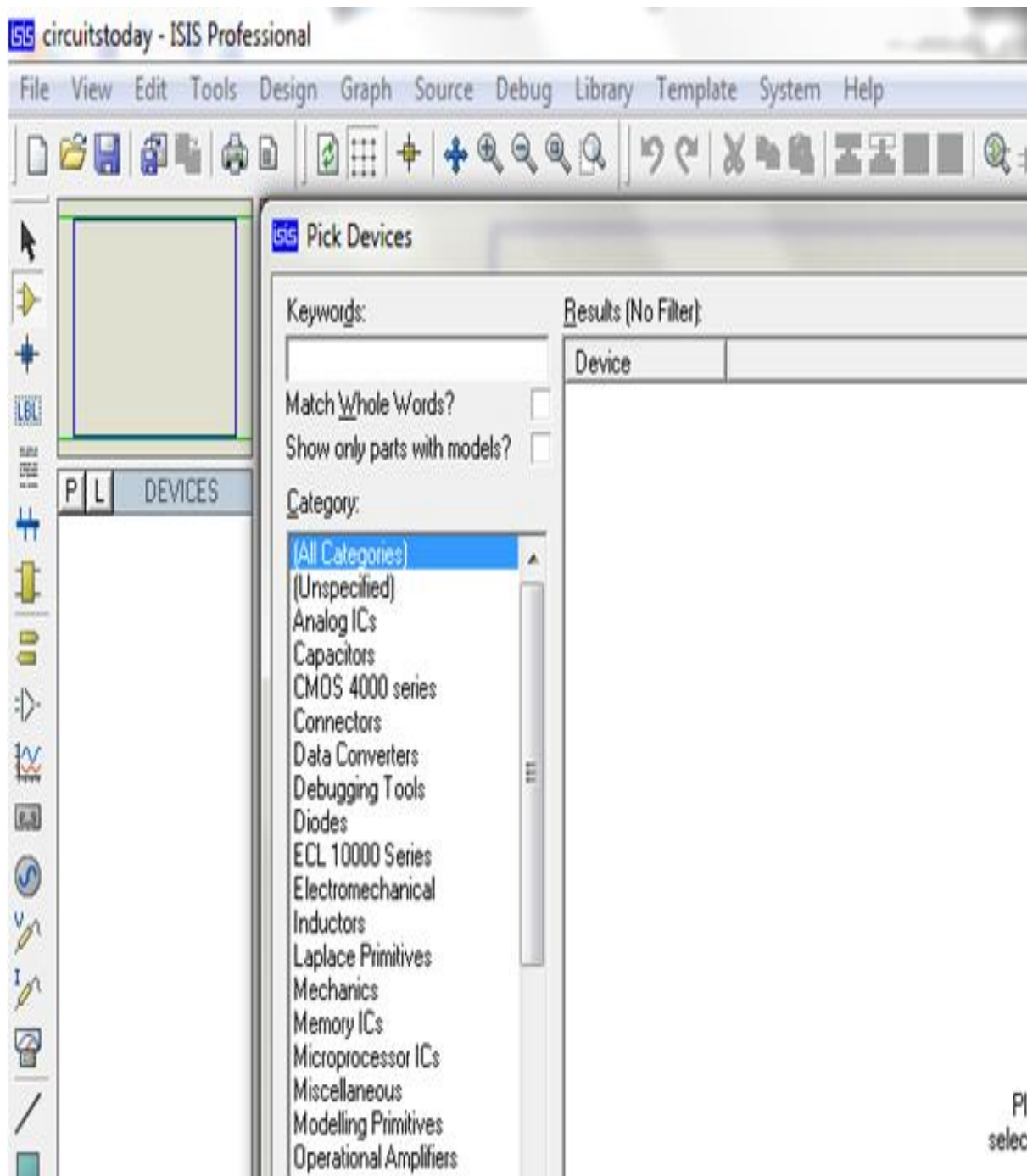
Component Mode

Step 5: Click On ARDUINO k from Libraries. It shows the categories of components available and a search option to enter the part name.



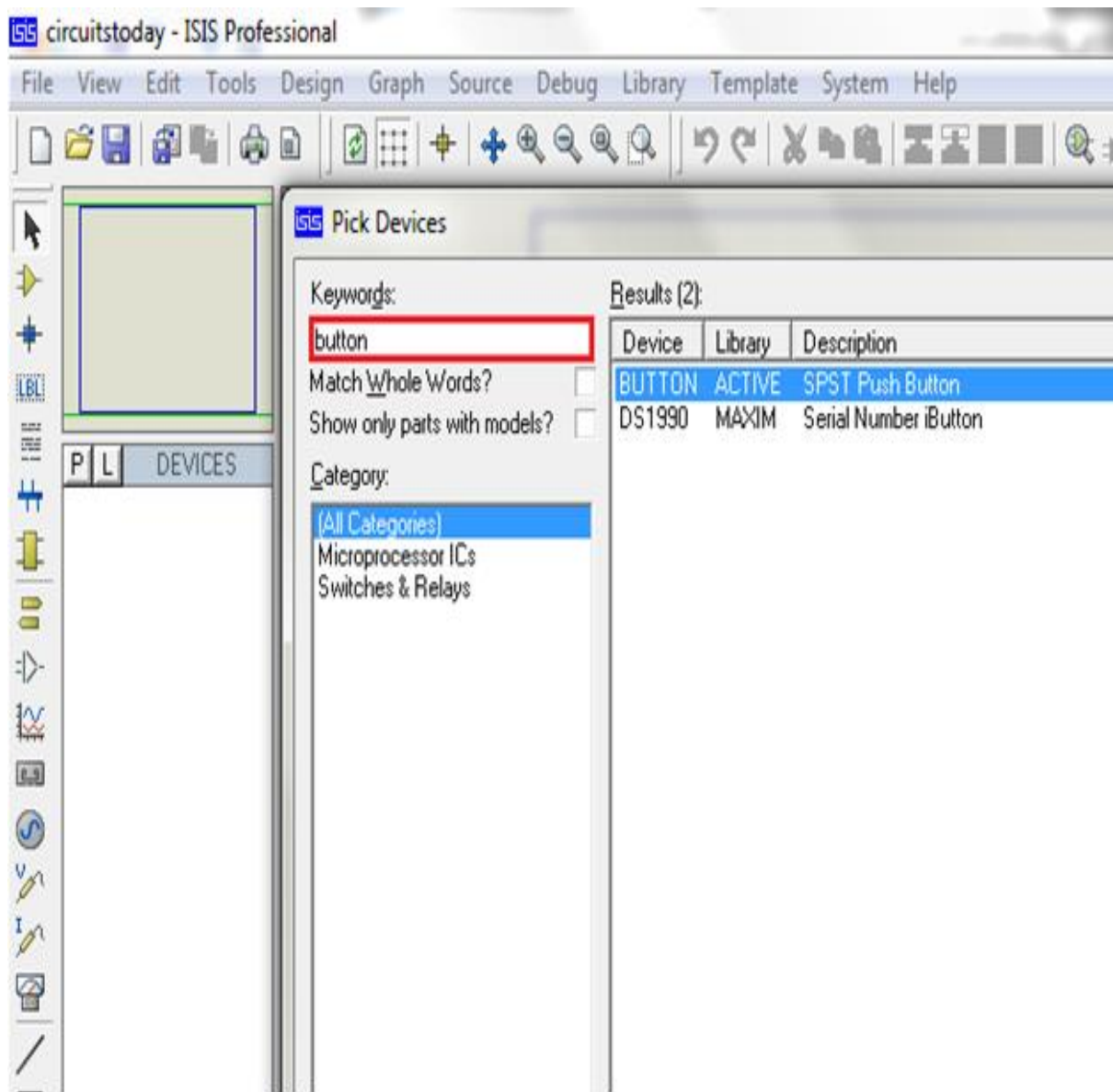
ARDUINOk from Libraries

Step 6: Select the components from categories or type the part name in Keywords text box.



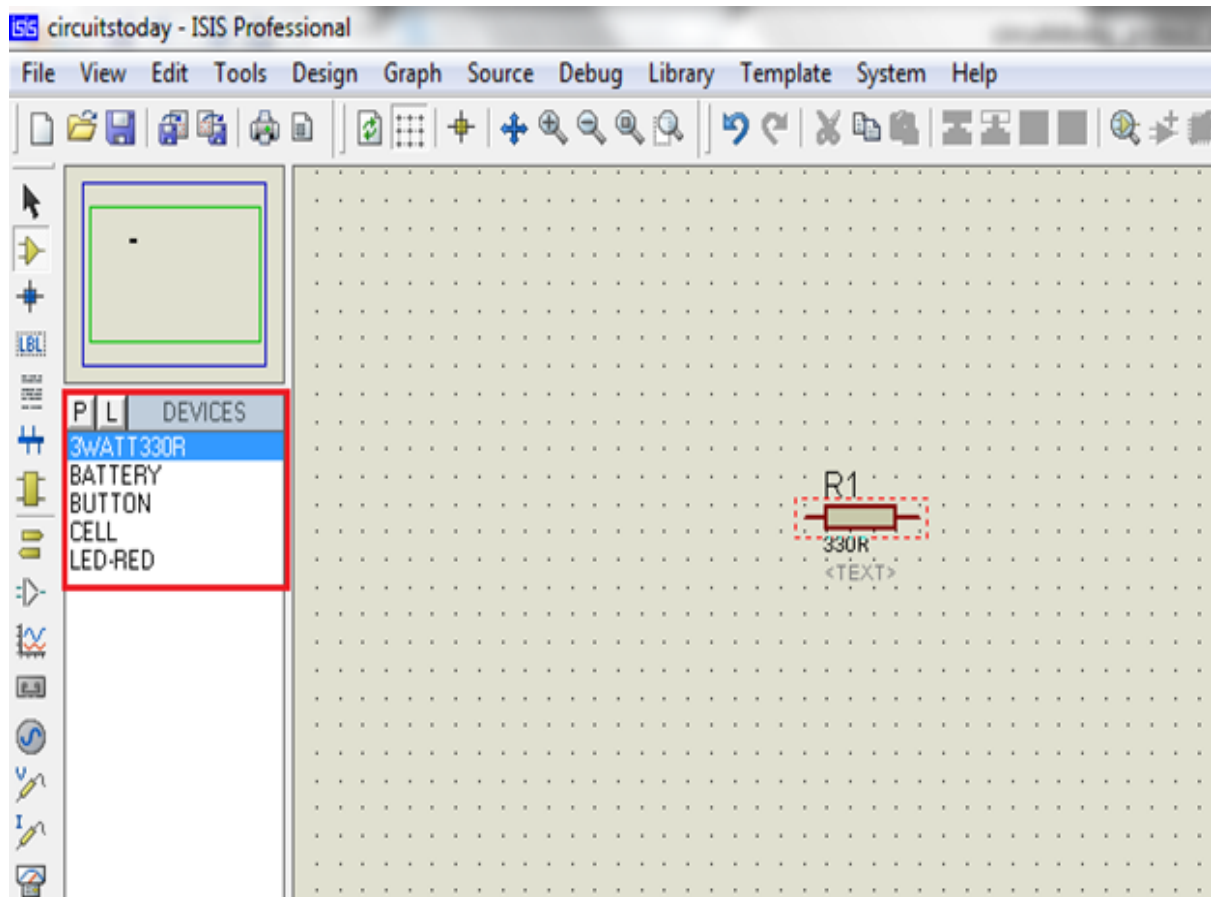
Keywords Textbox

Example shows selection of push button. Select the components accordingly.



Push Button Selection

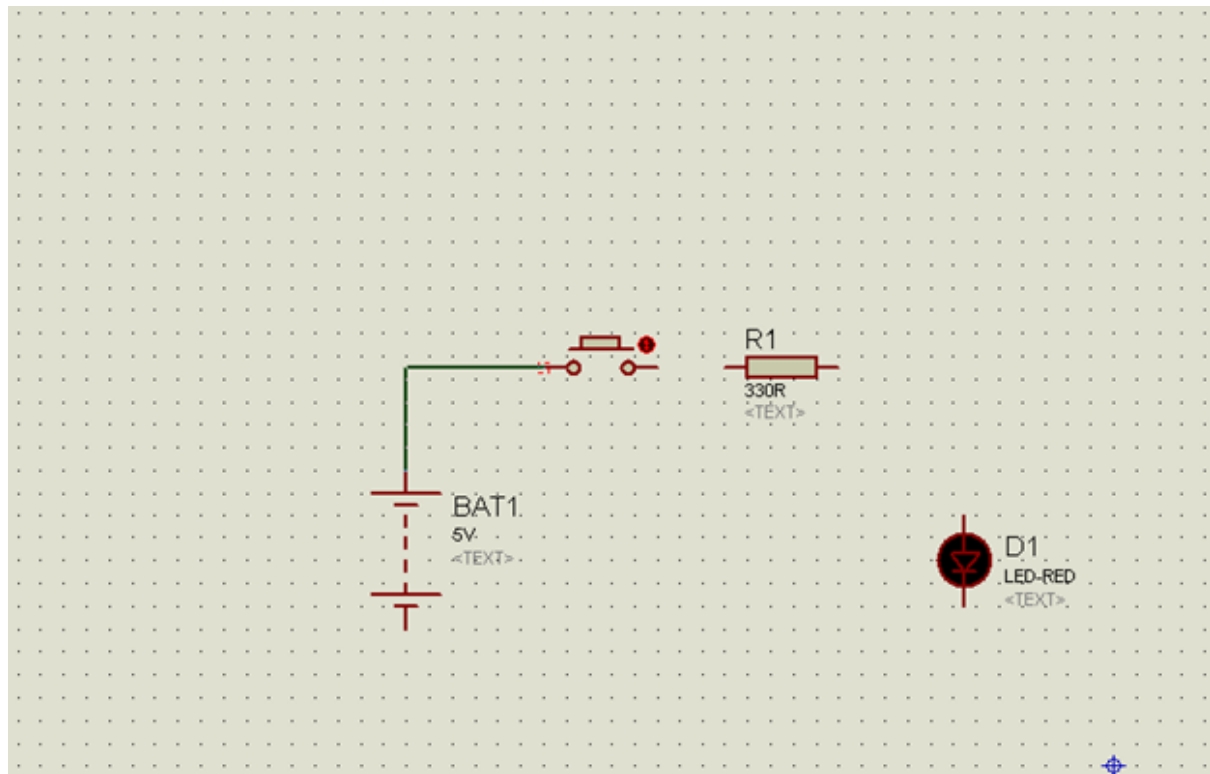
Step 7: The selected components will appear in the devices list. Select the component and place it in the design sheet by left-click.



Component Selection

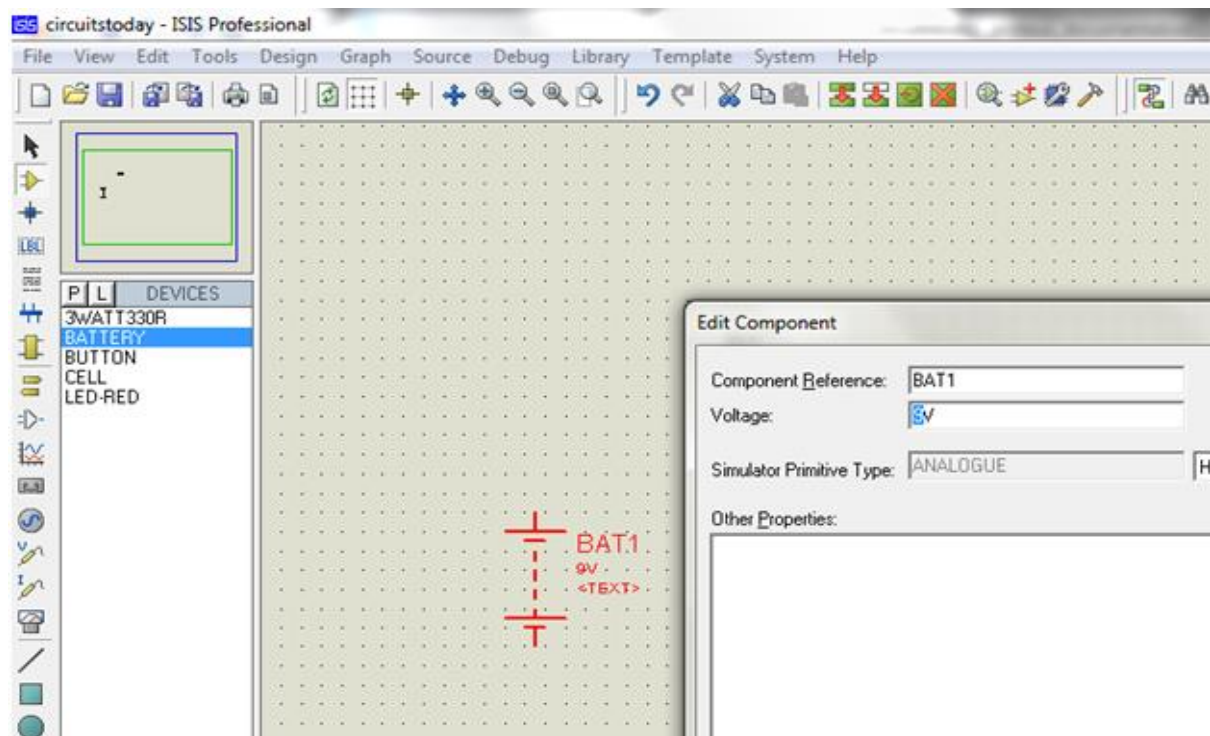
Place all the required components and route the wires i.e, make connections.

Either selection mode above the component mode or component mode allows to connect through wires. Left click from one terminal to other to make connection. Double right-click on the connected wire or the component to remove connection or the component respectively.



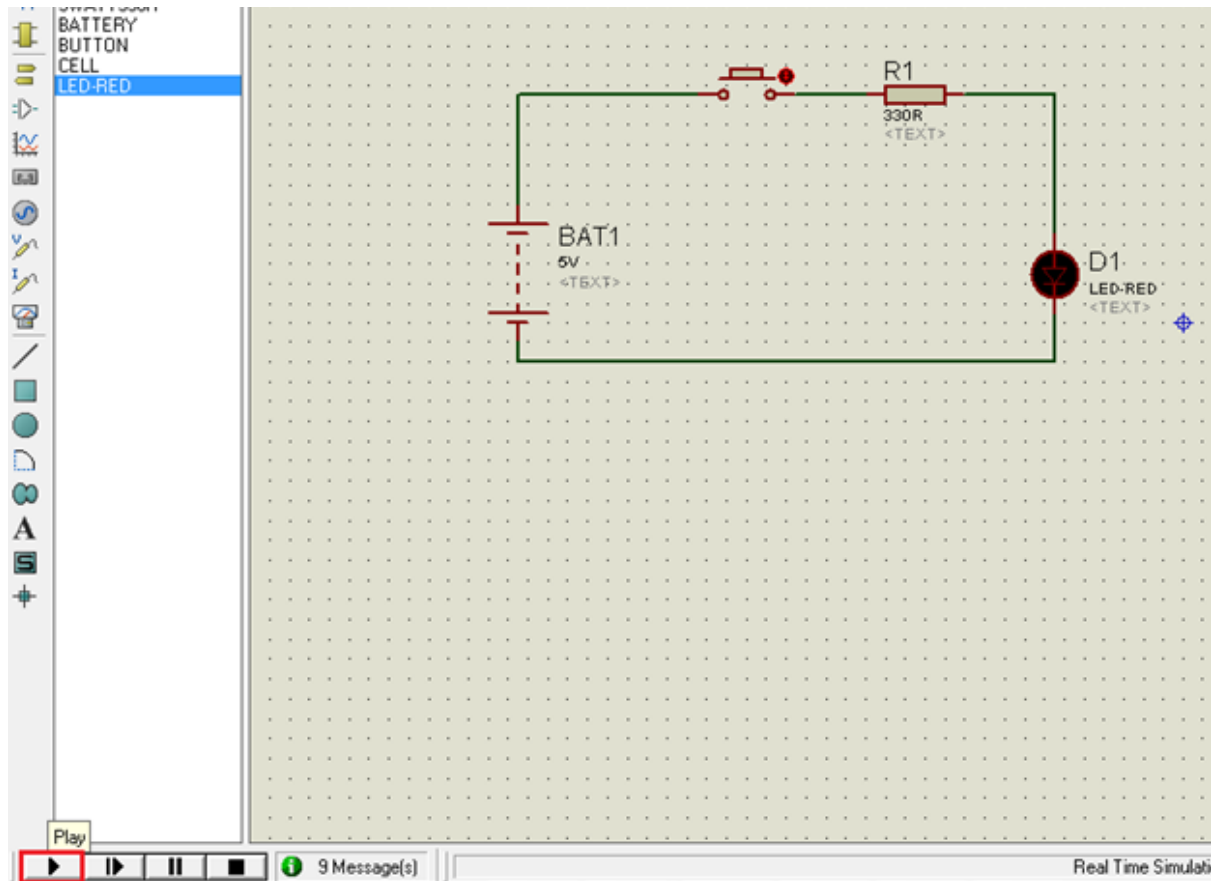
Component Properties Selection

Double click on the component to edit the properties of the components and click on Ok.



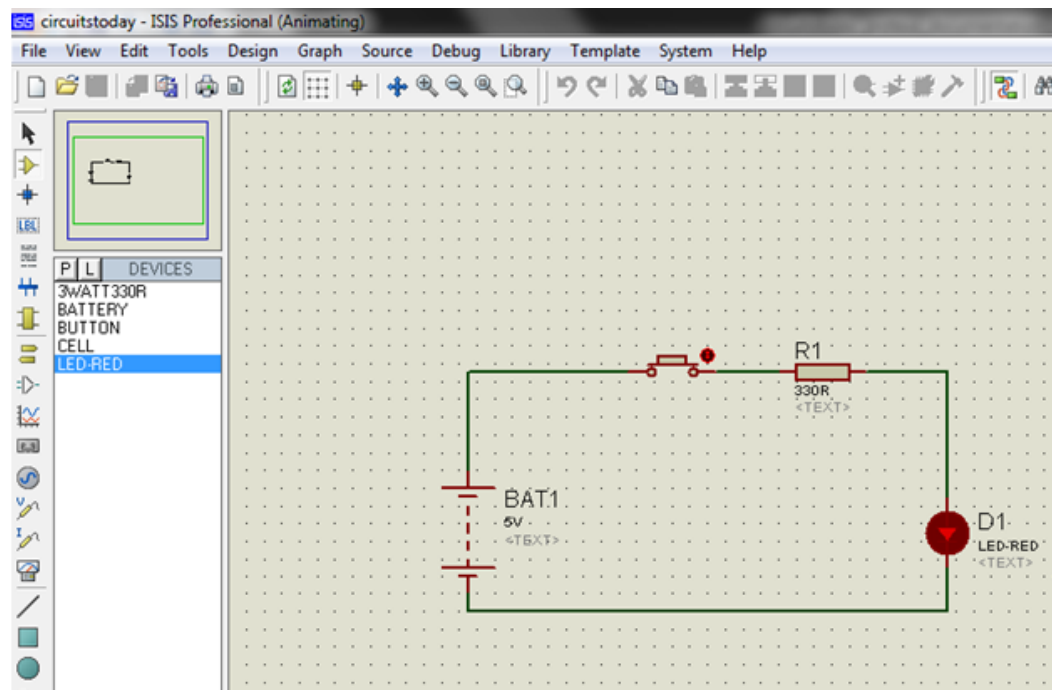
Component Properties Edit

Step 8:After connecting the circuit,click on the play button to run the simulation.



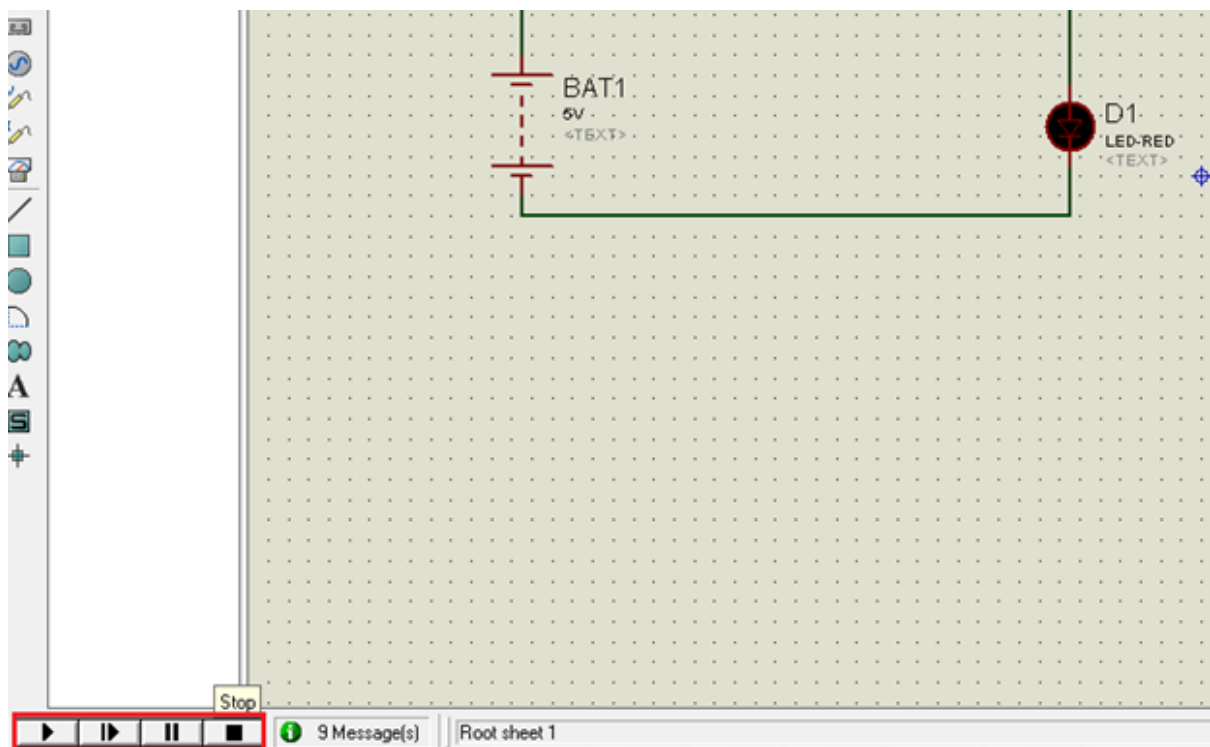
Simulation Run

In this example simulation, the button is depressed during simulation by clicking on it to make LED glow.



Simulation Animating

Simulation can be stepped, paused or stopped at any time.



Simulation Step-Pause-Stop Buttons

Step 1 – First you must have your Arduino board (you can choose your favorite board) and a USB cable. In case you use Arduino UNO, Arduino Duemilanove, Nano, Arduino Mega 2560, or Diecimila, you will need a standard USB cable (A plug to B plug), the kind you would connect to a USB printer as shown in the following image.

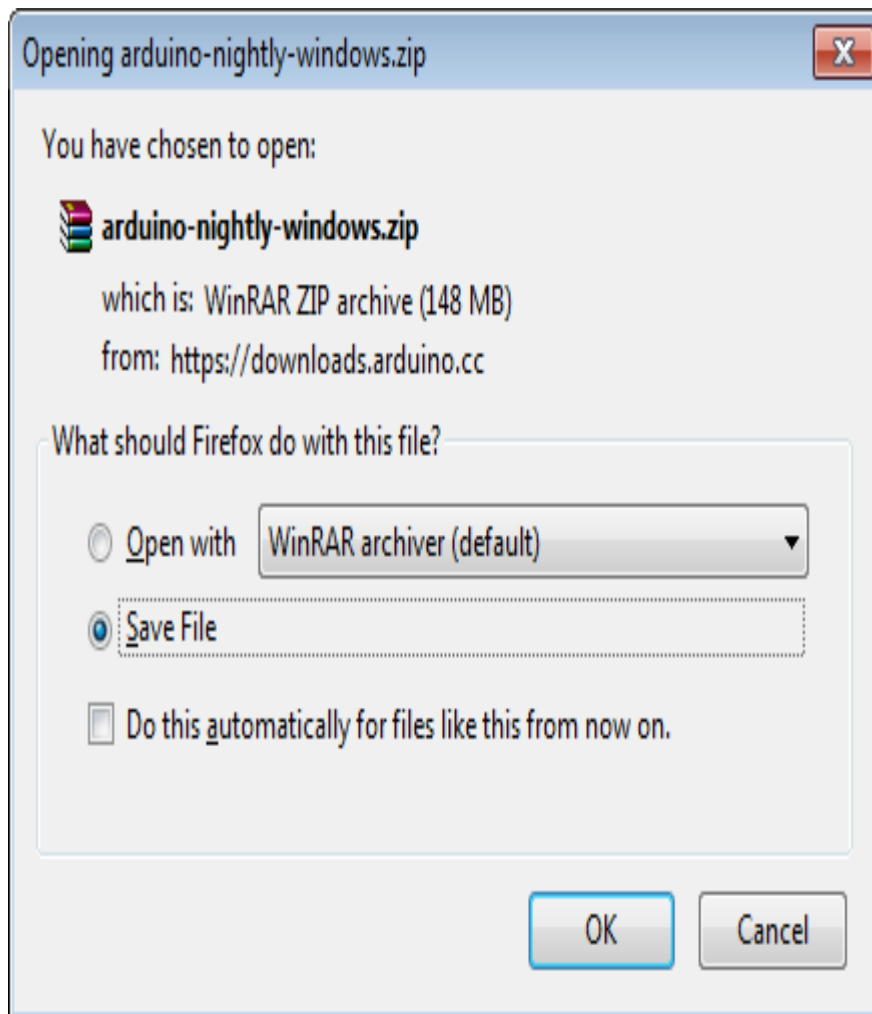


In case you use Arduino Nano, you will need an A to Mini-B cable instead as shown in the following image.



Step 2 – Download Arduino IDE Software.

You can get different versions of Arduino IDE from the [Download page](#) on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.



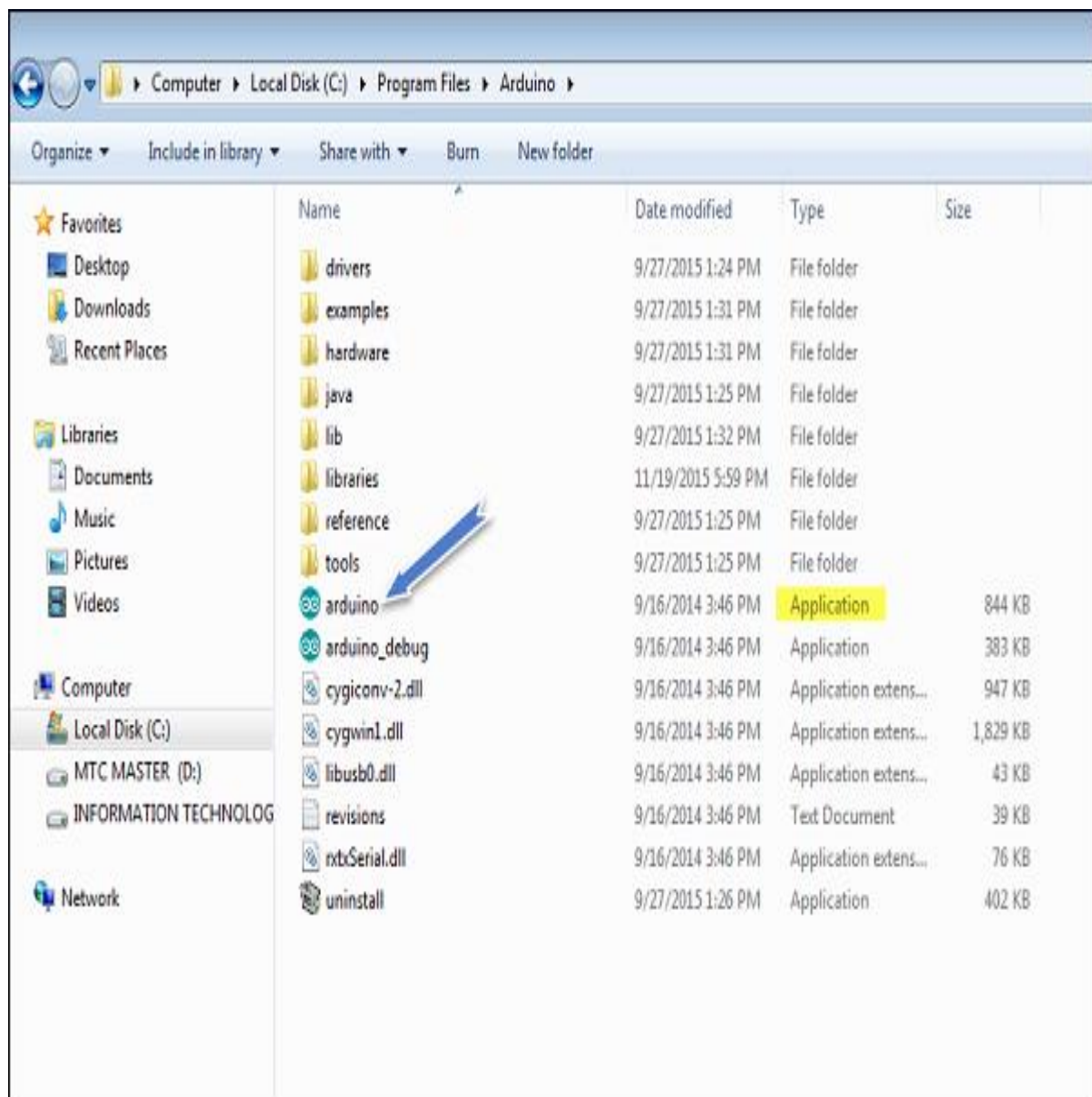
Step 3 – Power up your board.

The Arduino Uno, Mega, Duemilanove and Arduino Nano automatically draw power from either, the USB connection to the computer or an external power supply. If you are using an Arduino Diecimila, you have to make sure that the board is configured to draw power from the USB connection. The power source is selected with a jumper, a small piece of plastic that fits onto two of the three pins between the USB and power jacks. Check that it is on the two pins closest to the USB port.

Connect the Arduino board to your computer using the USB cable. The green power LED (labeled PWR) should glow.

Step 4 – Launch Arduino IDE.

After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Double-click the icon to start the IDE.

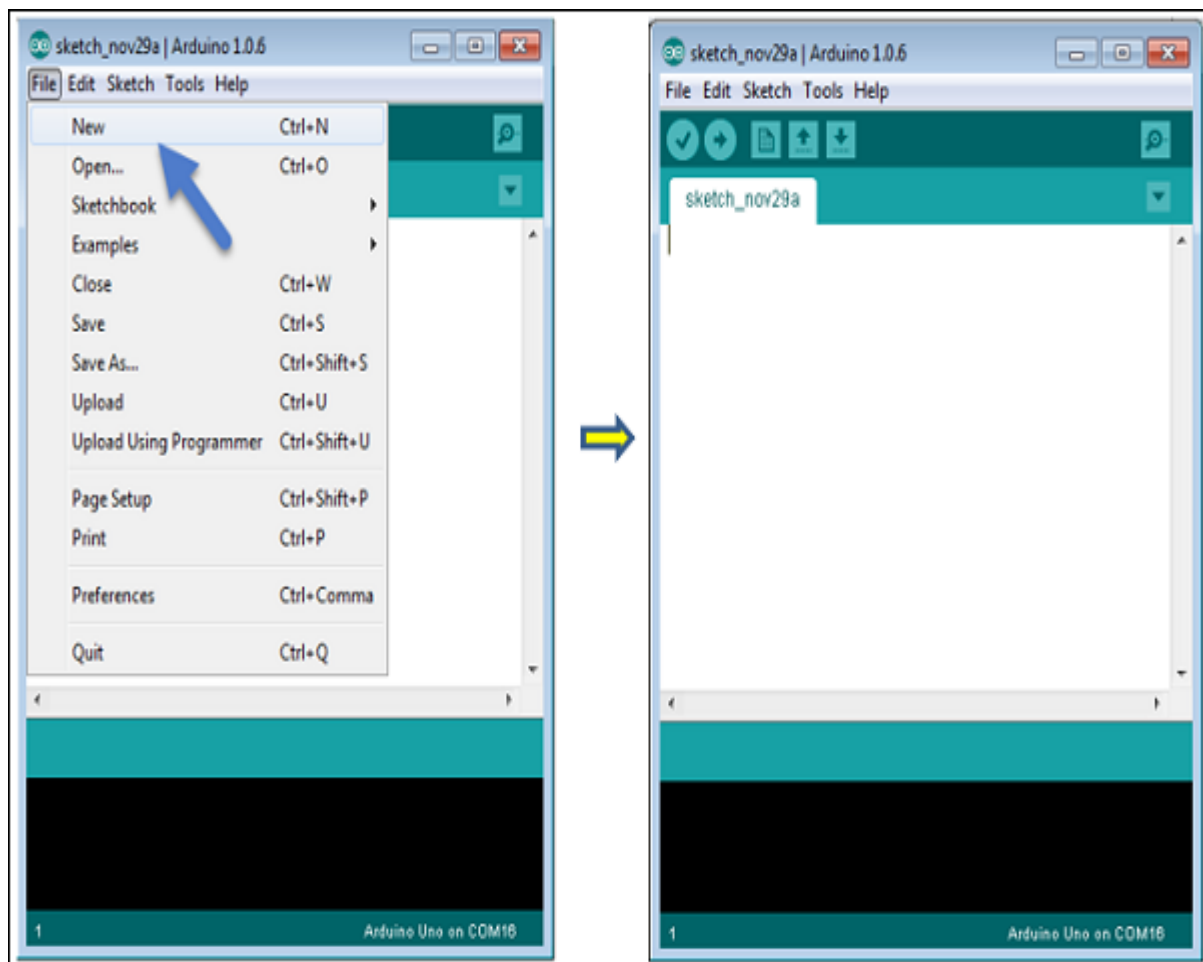


Step 5 – Open your first project.

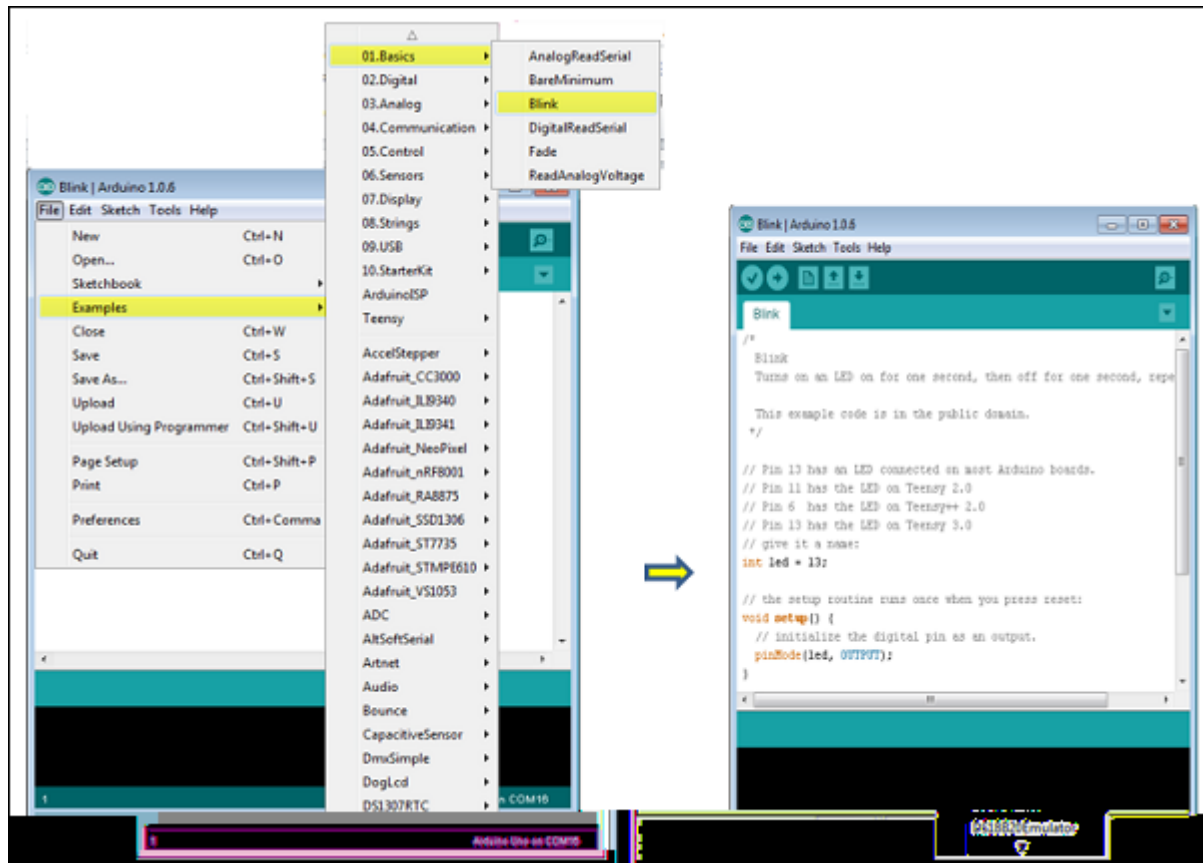
Once the software starts, you have two options –

- Create a new project.
- Open an existing project example.

To create a new project, select File → **New**.



To open an existing project example, select File → Example → Basics → Blink.

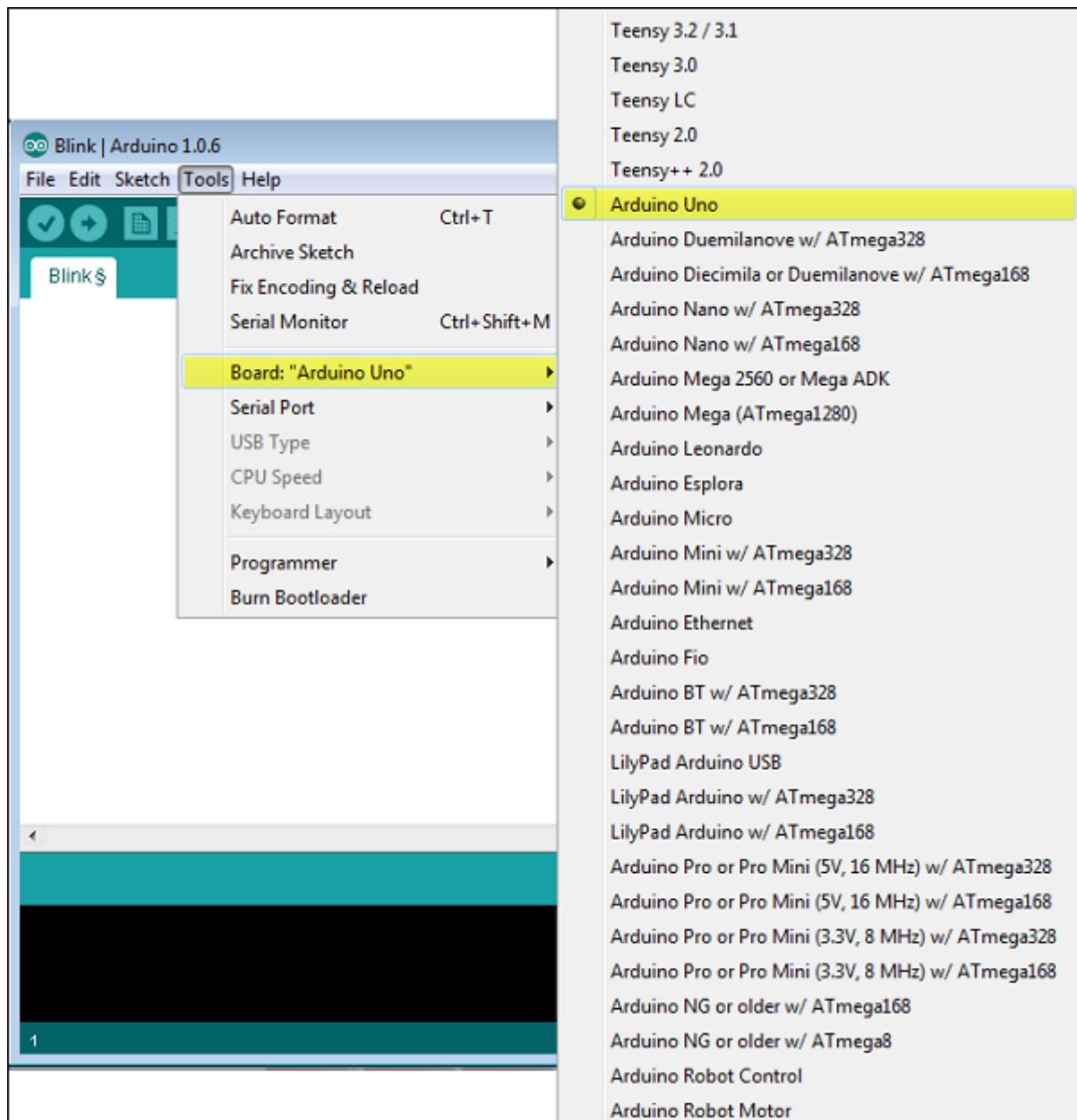


Here, we are selecting just one of the examples with the name **Blink**. It turns the LED on and off with some time delay. You can select any other example from the list.

Step 6 – Select your Arduino board.

To avoid any error while uploading your program to the board, you must select the correct Arduino board name, which matches with the board connected to your computer.

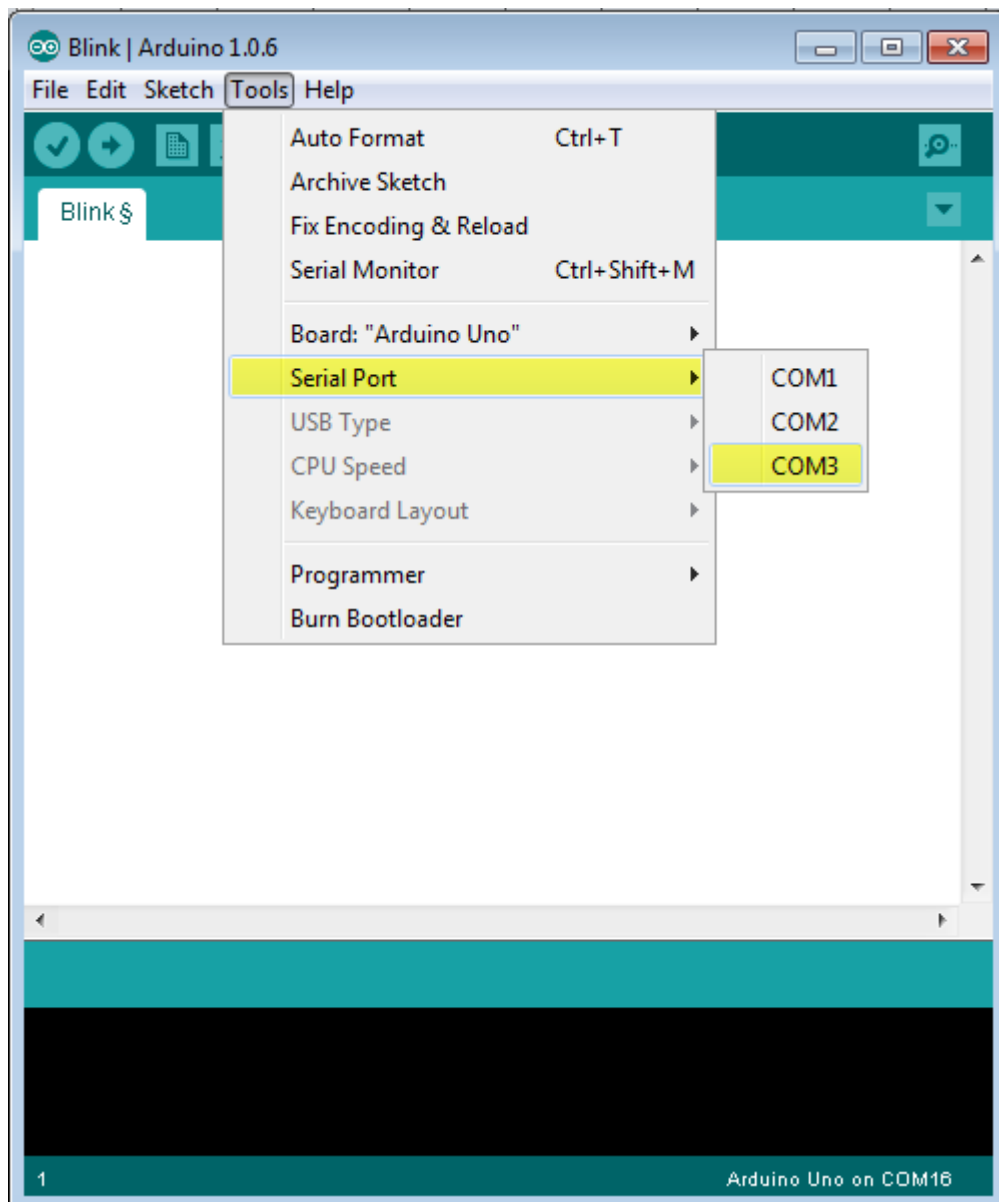
Go to Tools → Board and select your board.



Here, we have selected Arduino Uno board according to our tutorial, but you must select the name matching the board that you are using.

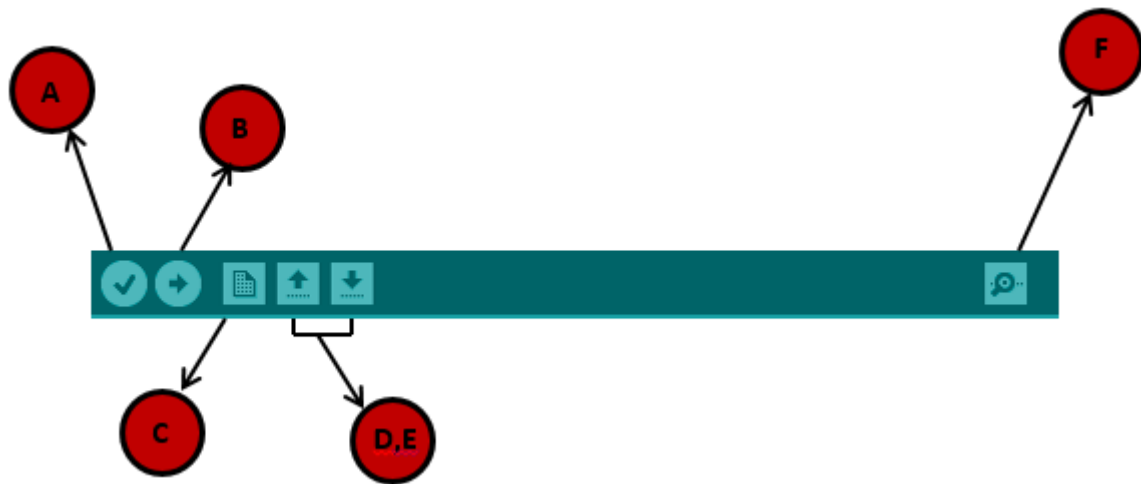
Step 7 – Select your serial port.

Select the serial device of the Arduino board. Go to **Tools** → **Serial Port** menu. This is likely to be COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports). To find out, you can disconnect your Arduino board and re-open the menu, the entry that disappears should be of the Arduino board. Reconnect the board and select that serial port.



Step 8 – Upload the program to your board.

Before explaining how we can upload our program to the board, we must demonstrate the function of each symbol appearing in the Arduino IDE toolbar.



A – Used to check if there is any compilation error.

B – Used to upload a program to the Arduino board.

C – Shortcut used to create a new sketch.

D – Used to directly open one of the example sketch.

E – Used to save your sketch.

F – Serial monitor used to receive serial data from the board and send the serial data to the board.

Now, simply click the "Upload" button in the environment. Wait a few seconds; you will see the RX and TX LEDs on the board, flashing. If the upload is successful, the message "Done uploading" will appear in the status bar.

4.4 Program code:

```
#include <Servo.h>

Servo servo0;
Servo servo1;
Servo servo2;
Servo servo3;

int sensorvalue0;
int sensorvalue1;
int sensorvalue2;
int sensorvalue3;

void setup()
{
    pinMode(A0,INPUT);
    pinMode(3,OUTPUT);
    servo0.attach(3);

    pinMode(A1,INPUT);
    pinMode(5,OUTPUT);
    servo1.attach(5);

    pinMode(A2,INPUT);
    pinMode(6,OUTPUT);
    servo2.attach(6);

    pinMode(A3,INPUT);
    pinMode(9,OUTPUT);
    servo3.attach(9);
}
```

```
void loop()
{
  sensorvalue0 = analogRead(A0);
  sensorvalue0 = map(sensorvalue0, 0, 1023, 0, 180);
  servo0.write(sensorvalue0);
  sensorvalue1 = analogRead(A1);
  sensorvalue1 = map(sensorvalue1, 0, 1023, 0, 180);
  servo1.write(sensorvalue1);
  sensorvalue2 = analogRead(A2);
  sensorvalue2 = map(sensorvalue2, 0, 1023, 0, 180);
  servo2.write(sensorvalue2);
  sensorvalue3 = analogRead(A3);
  sensorvalue3 = map(sensorvalue3, 0, 1023, 0, 180);
  servo3.write(sensorvalue3);
}
```

CHAPTER 5

Conclusion and Future Scope

5.1 Conclusion

Robots are examples of programmable automation, however they are also used in exible or even xed automation systems. The two reasons for selecting a robot to operate in a production line are rst to reduce labour costs and second to perform work that is boring, unpleasant or hazardous for human beings. Robot can perform repetitive tasks at a steady place, be programmed to achieve and perform different unpleasant tasks, operation for long period without rest or break period, and response in automation manufacturing operation on a continuous basis.

Robot technology is an applied science that is referred to as a combination of machine tool fundamentals and computer application, the variety of technical features about the way a robot is construction and works and the factors that influence its selection robot system are usually classified as low technology and high technology groups. The capacity of a robot to position and oriented the end of its arm with accuracy and repeatability is an important control attribute. Among the important properties of a robot to properly regulate its motion are :

Stability

Control Resolution

Spatial Resolution

Accuracy

Repeatability

Compliance

Programming is the process of preparing a detailed sequence of operating instruction to solve a particular problem, testing it to ensure its accuracy, and preparing documentation to be run on a digital computer.

Robot programming can be defined as a path in space through which the manipulator is directed to move. A program is a logically arranged set of programming instructions. The capabilities of a robot operating system depend on the type of controller, the arm configuration and the method used to train the robot.

5.2 Future Scope

Can be used to interact with the household objects and observing the result with the help of camera system.

Can be used for accurate holding and inspection applications in the industries.

It can be used for a pick and place applications in industries like soap manufacturing, pharmacy etc.

The challenge for the future has been analysed and evaluated because more effort has gone into developing automated systems to improve productivity than has gone into the appropriate matching of people and technology.

The versatility of the robots used in all the applications is important, as in applying that capability efficiently. Future application of robots have been explored, and critical areas have been analysed.

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